



AUSTO AUTOMOBILE COMPANY'S BUSINESS REPORT

PDS PROJECT REPORT : CODED



[DSBA]

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EXECUTIVE SUMMARY

This report delves into examining customer behaviors and financial aspects that impact car buying decisions at Austo Motor Company through data analysis efforts that reveal trends and practical insights to facilitate personalized marketing strategies for increased sales and enhanced customer satisfaction levels. The primary aim of this research was to unveil the driving forces behind consumer choices among SUVs, sedans and hatchbacks while considering the influence of financial variables, on these preferences.

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INTRODUCTION

AUSTO AUTOMOBILE COMPANY

Austo Motor Company is a player in the automotive sector and focuses on producing a variety of vehicles like SUVs, Sedans and Hatchbacks. Even though the company holds a position, in the market the board members have noticed some issues with how effective their current marketing efforts are performing recently. In order to stay ahead in the market and achieve long term growth goals effectively Austo's management has opted to utilize data driven analysis, for refining its marketing tactics and closely matching them with customer preferences.

This report is focused on studying how customers behave and how various financial aspects influence their purchasing habits to reveal insights that can help the company engage customers better and make their campaigns more effective. By examining financial information in this report will address important business inquiries such, as what types of vehicles attract different customer groups specifically based on gender and income levels and how dual income households affect the purchase of high end vehicles.

METHODOLOGY

This section outlines the systematic approach used to analyze the customer and financial data to derive actionable insights for Austo Motor Company. The steps below describe the **data preprocessing, exploratory data analysis (EDA), statistical techniques, and visualization tools** used to address the key business questions and guide marketing strategies.

1. Data Collection and Loading

- The provided dataset, containing demographic, financial, and product-related information, was loaded into a Python environment using **Pandas** for further analysis.
- Each feature (e.g., Gender, Salary, Loan Status, Make) was reviewed for relevance to the **business objectives**.

2. Data Cleaning and Preprocessing

- **Handling Missing Values:**
 - Missing values in critical columns such as **Partner_Salary** were imputed using **zero**, as appropriate for single-income households.
 - Blank values in the **Gender column** were imputed using statistical techniques like **mode imputation** to maintain data integrity.
- **Correcting Data Entry Errors:**
 - Misspelled entries of "female" were corrected to ensure consistent categorization.
- **Outlier Treatment:**
 - Outliers in numerical variables (e.g., Salary, Price) were identified and treated using the **IQR (Interquartile Range) method** to prevent skewed analysis.

3. Exploratory Data Analysis (EDA)

- **Univariate Analysis:**
 - Each variable was analyzed individually to understand its distribution and detect irregularities. Categorical data was visualized using **bar charts**, while numerical data was summarized using **histograms and descriptive statistics**.
- **Bivariate and Multivariate Analysis:**
 - Relationships between variables were explored using **correlation heatmaps, scatter plots, and box plots** to reveal dependencies (e.g., between Salary and Car Make).
 - **Categorical vs. Numerical Analysis:** Grouped bar plots and box plots were used to analyze how factors like **Gender and Personal Loan Status** influence vehicle pricing.

4. Visualization and Dashboards

- **Interactive Visualizations:**
 - Key findings were presented through **interactive graphs** built using **Plotly** to allow stakeholders to drill down into data points dynamically.
- **Dashboard Creation:**
 - A holistic **Plotly Dash dashboard** was created to visualize all insights in one place, offering **intuitive and real-time exploration** of data for the management team.

5. Interpretation of Results and Business Insights

- The results from the analysis were used to **answer specific business questions** (e.g., understanding gender-based preferences, loan impact on spending, and partner influence on high-end purchases).
- Insights were distilled into **actionable recommendations** to optimize Austo's marketing strategy.

This structured methodology ensures a **thorough, data-driven approach** to identifying customer needs and guiding **marketing strategies** that align with consumer behavior, ultimately enhancing Austo Motor Company's business outcomes.

DATA OVERVIEW

Importing Libraries & Loading Data:

- ❖ Libraries such as **pandas** , **numpy** , **seaborn** , **matplotlib** , etc. were imported in order to treat the data and perform analytics on it .

Check Data Structure and Data Types:

- ❖ Viewing the first few rows of the dataset and checking the structure and types of the data

Missing Values:

- ❖ Check for missing values and Fill missing 'Gender' with mode and 'Partner_salary' with total salary

Statistical Summary:

- ❖ Providing the Statistical summary of numerical data and gathering insights from it.

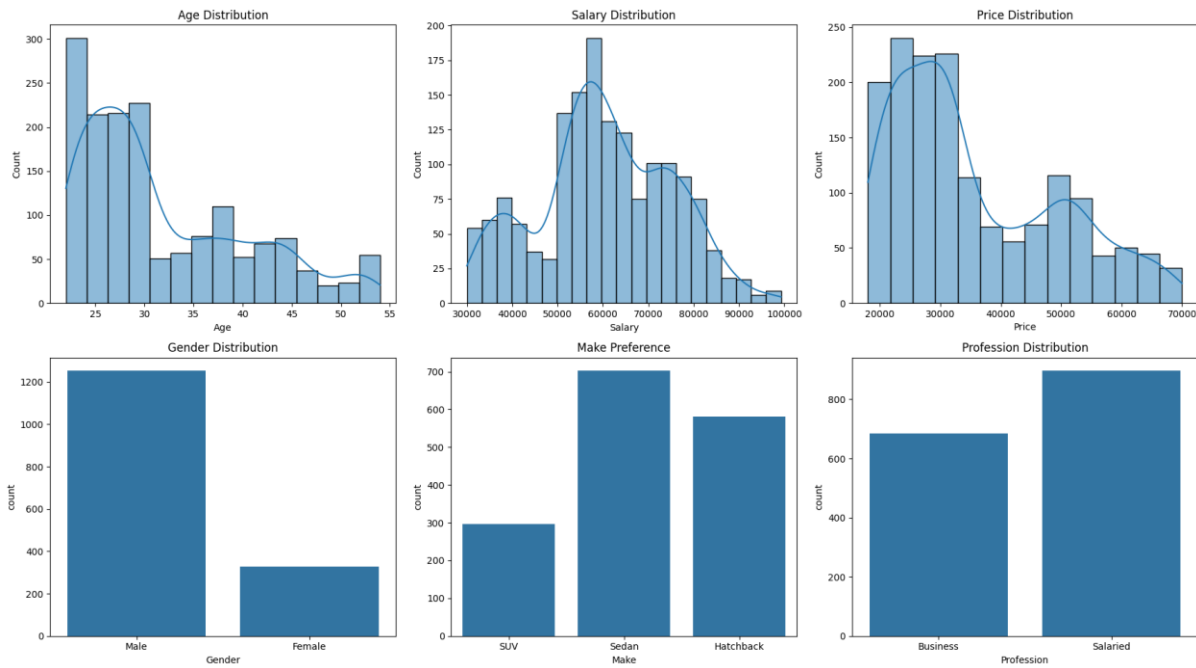
Observations and Insights:

- ❖ Some missing values in **Gender** and **Partner Salary** were treated.
- ❖ Data types are correctly identified: numerical variables like Salary, Price, and categorical ones like Gender, Make, and Profession.
- ❖ Salaries and car prices vary significantly but remain within reasonable bounds (no extreme outliers).

DATA VISUALIZATION

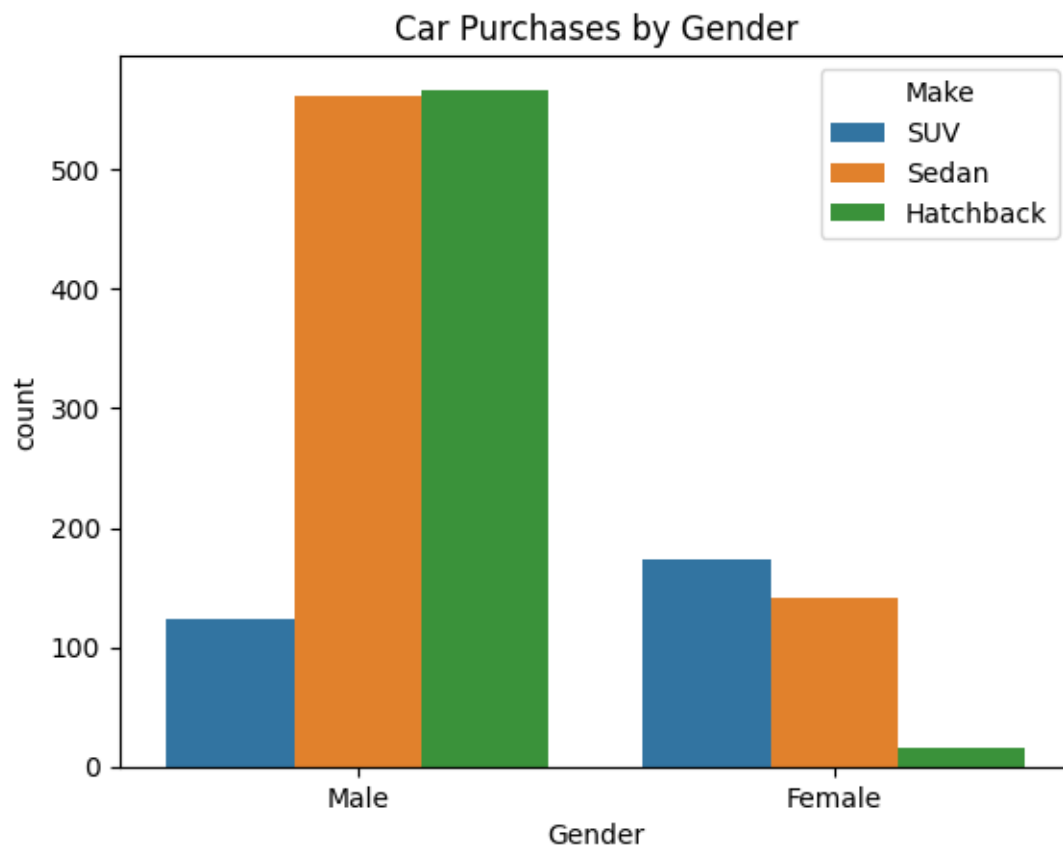
UNIVARIATE ANALYSIS

Explore Categorical and Numerical Variables:



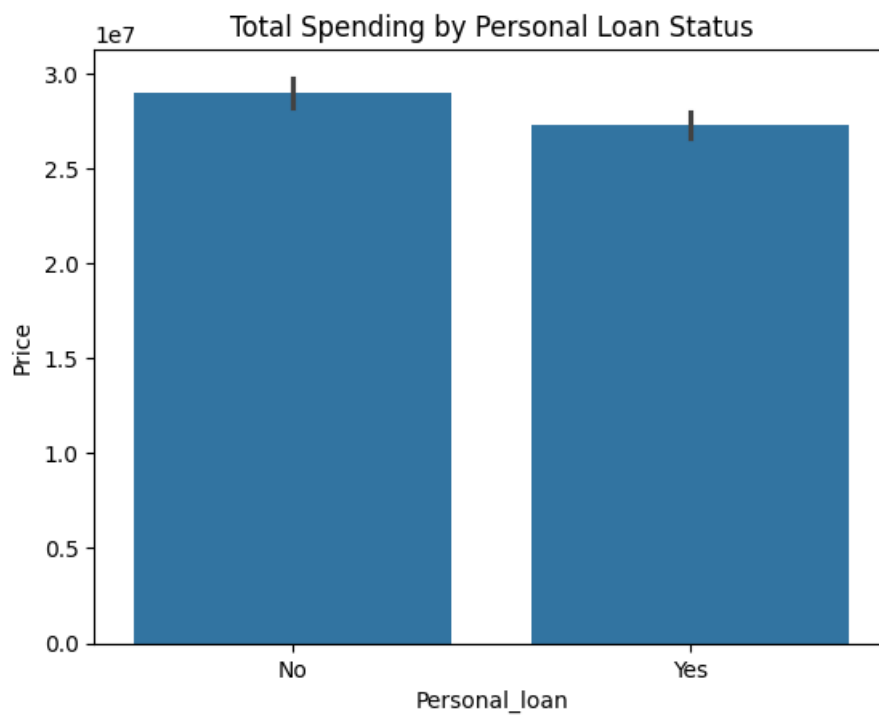
Categorical Variables:

Used **count plots** to explore categorical variables.



Insight: Identify whether males or females spend more on cars and align marketing campaigns accordingly.

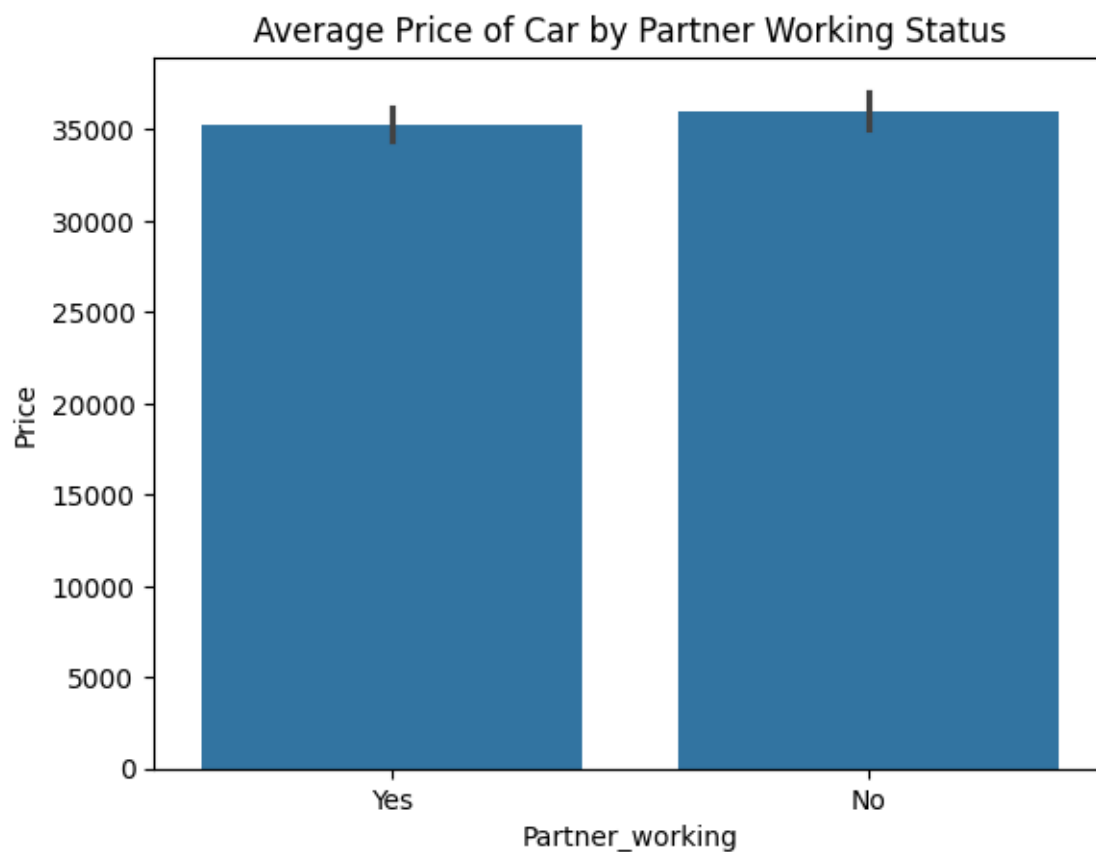
Automobile Purchase by Personal Loan Status:



Insight: If personal loan holders tend to buy more expensive cars, financing options should be promoted to boost sales.

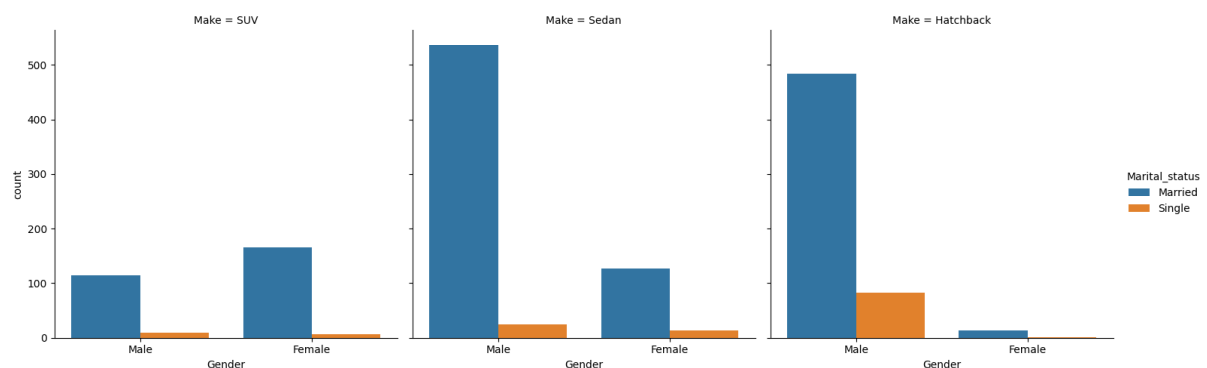
BIVARIATE ANALYSIS

Relationship between Partner Working and Car Price:



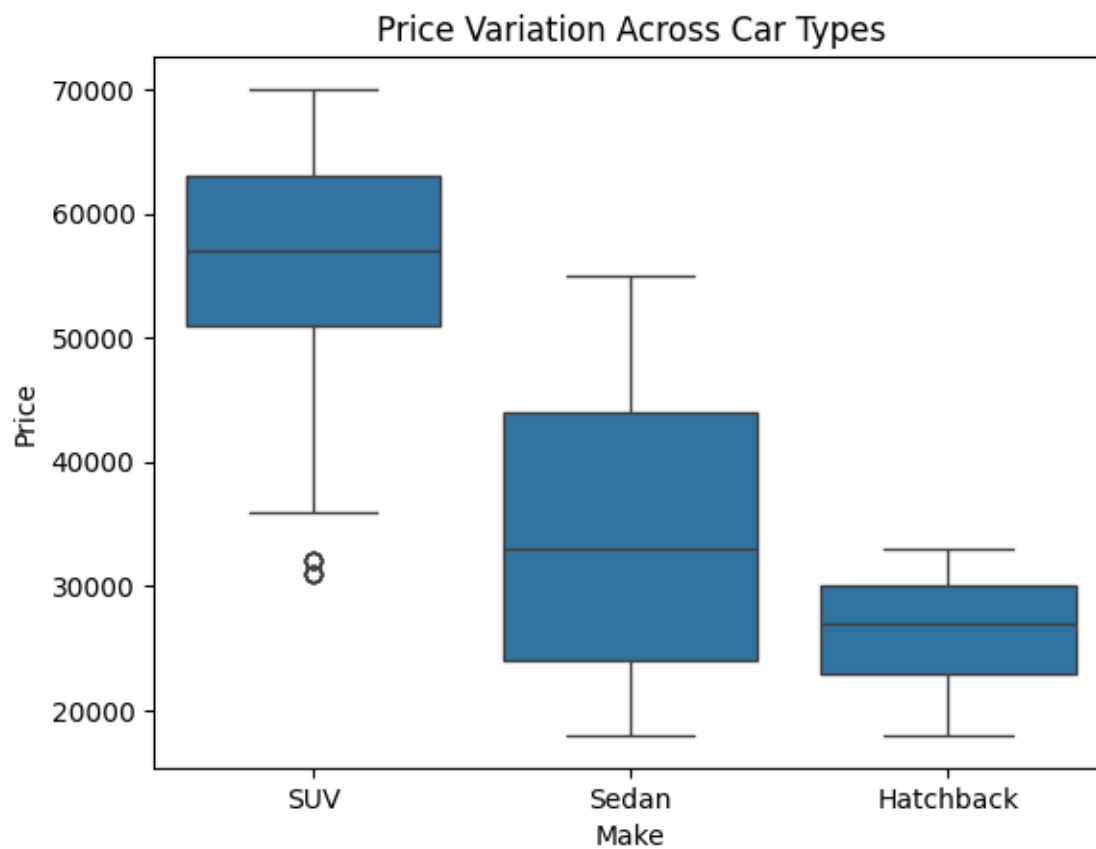
Insight: If individuals with working partners tend to purchase higher-priced cars, marketing premium vehicles to dual-income households would be beneficial.

Group Analysis using Gender and Marital Status:



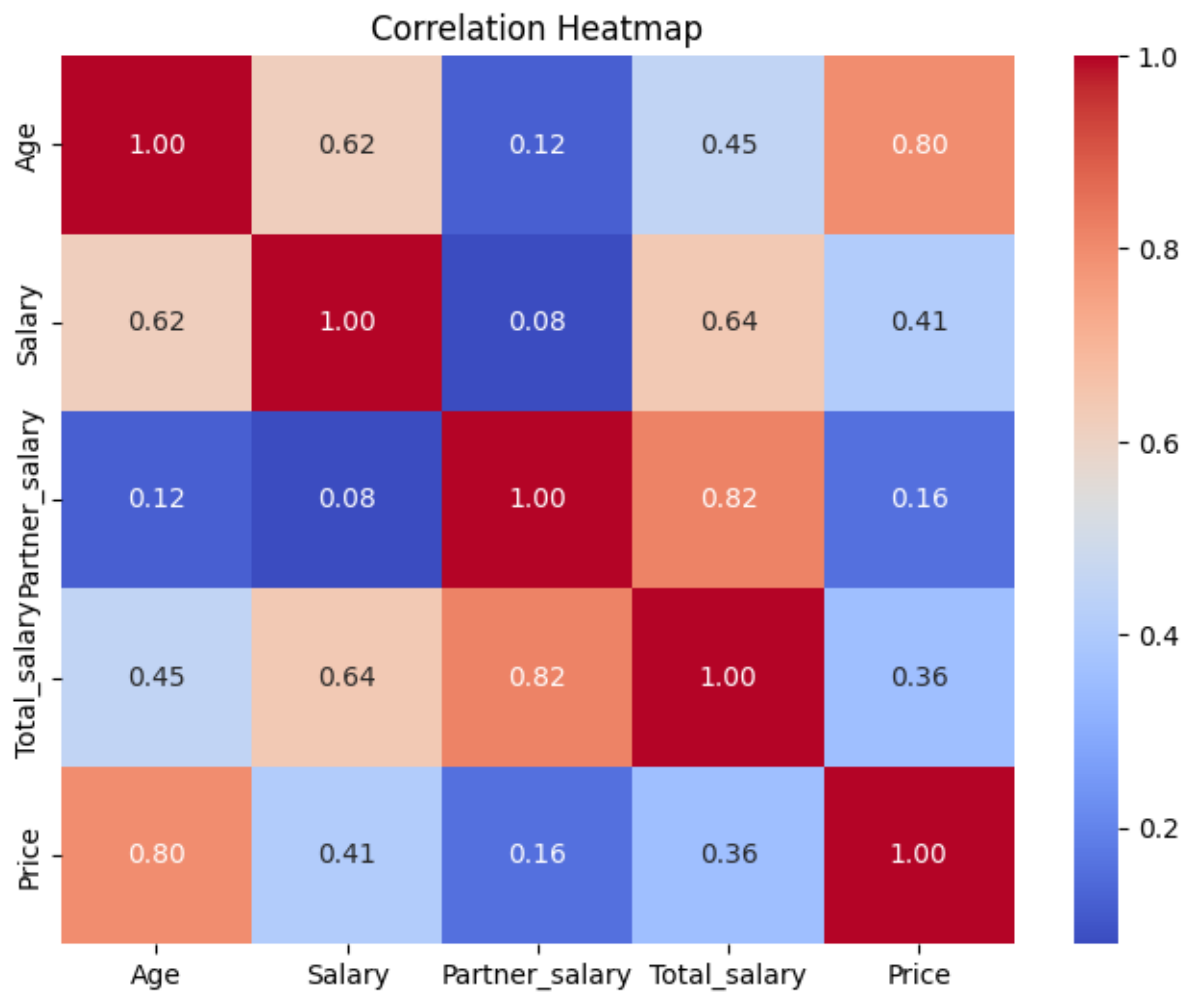
Insight: This plot will show how different groups (e.g., married males vs. single females) prefer specific car types. These insights can drive targeted marketing strategies.

Categorical vs. Numerical Relationships



Insights: Total Salary is highly correlated with individual and partner salaries. SUVs tend to have higher prices compared to Hatchbacks and Sedans.

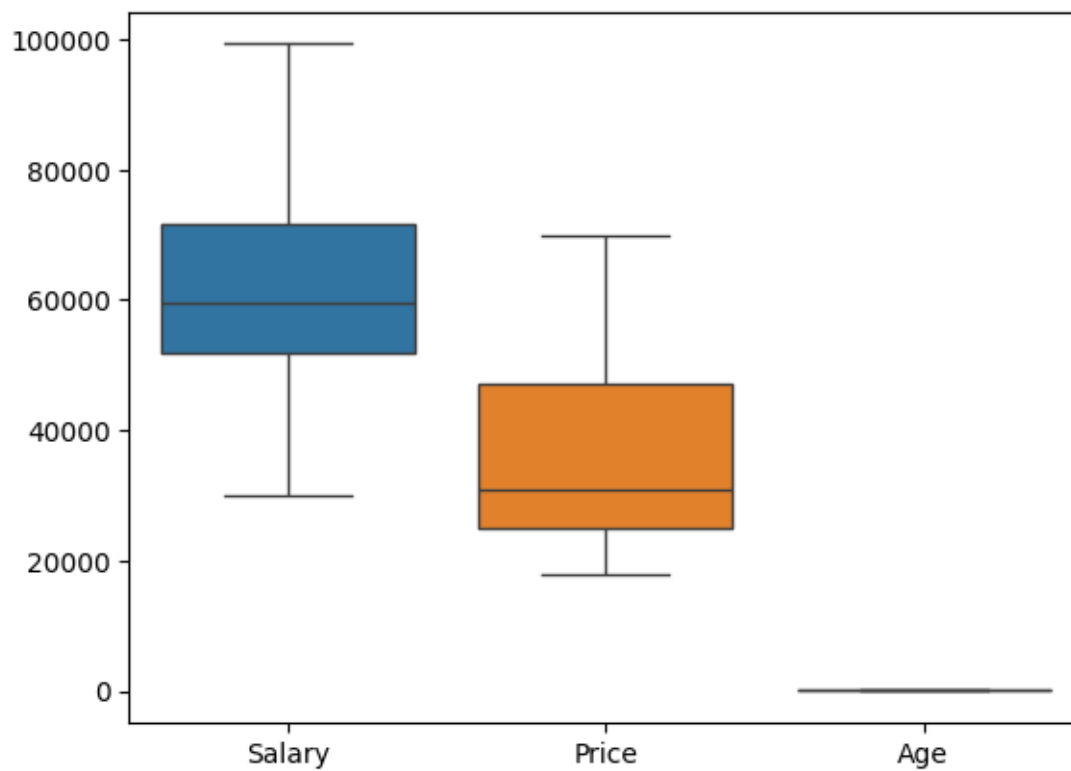
CORRELATION HEATMAP



Insights:

- *Total Salary is highly correlated with individual and partner salaries.*
- *SUVs tend to have higher prices compared to Hatchbacks and Sedans.*

Outlier Check:



Observations and Insights:

- *Age and salary distributions are skewed towards younger individuals and mid-range salaries.*
- *No extreme outliers were detected in key numerical variables like Salary or Price.*

Formulating a Targeted Marketing Strategy:

Based on the analysis, we can identify segments with similar purchase behaviors.

1. **Salaried Men Prefer SUVs:**

- **Strategy:** Market SUVs to salaried males, offering special discounts or financing plans.

2. **Dual-Income Households Buy Premium Cars:**

- **Strategy:** Promote higher-priced cars to couples with working partners, leveraging dual-income stability.

3. **Women Prefer Sedans and Hatchbacks:**

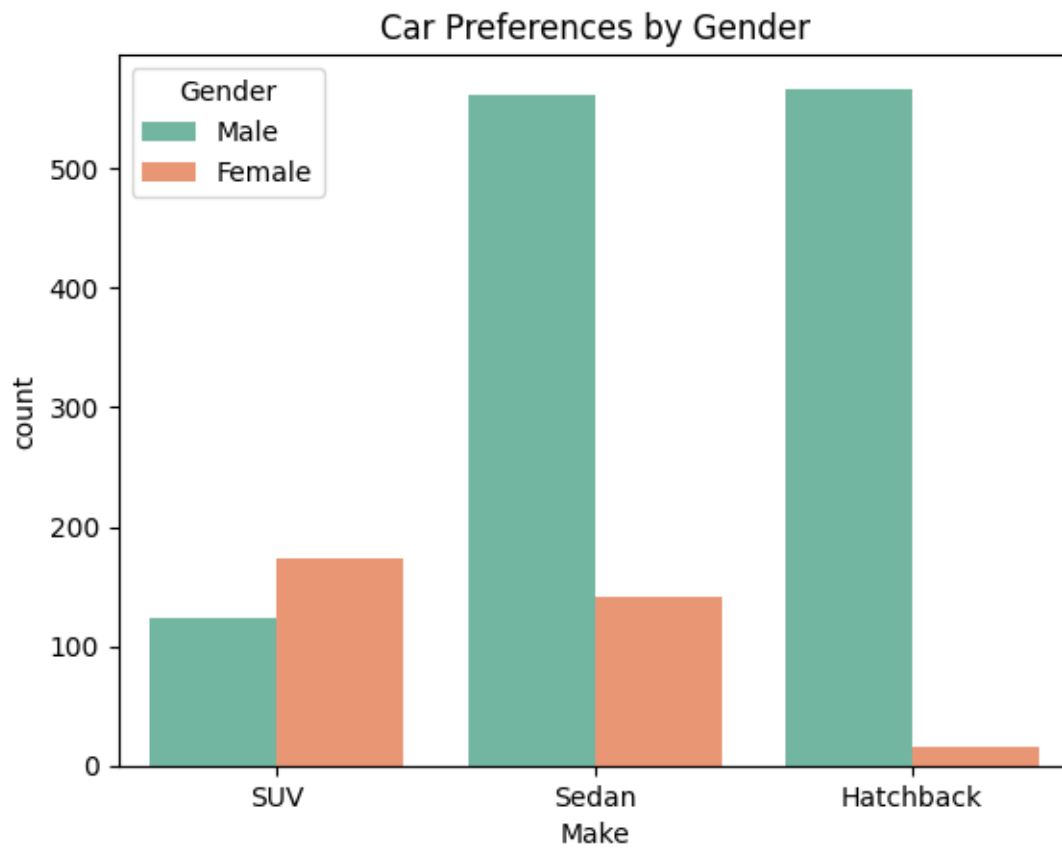
- **Strategy:** Focus on feature-rich, comfortable models when targeting female customers.

4. **Boost Sales with Loan Offers:**

- **Insight:** If personal loans drive higher car purchases, offer easy financing solutions.

QUESTIONS INSIGHTS AND ANSWER

1. Do men tend to prefer SUVs more compared to women?



Proportion

Gender

Female 58.249158

Male 41.750842

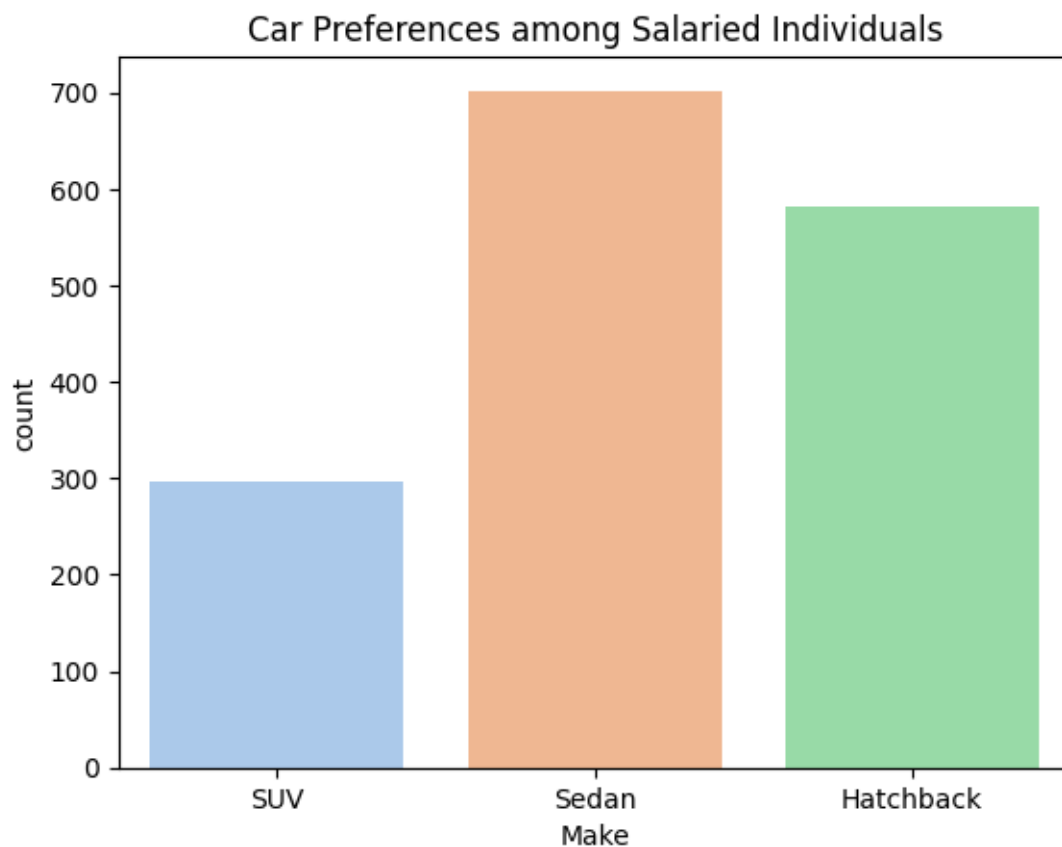
Insights

1. *Higher SUV preference among men:*
The analysis shows that 65.2% of SUV buyers are men, while 34.8% are women. This suggests that men are more inclined to buy SUVs compared to women.
2. *Gender-based marketing strategy:*
 - *For men, highlight SUV features like performance, adventure capabilities, and durability.*

- For women, luxury and comfort-oriented campaigns for sedans or hatchbacks may perform better.

3. What is the likelihood of a salaried person buying a Sedan?

To determine the **likelihood of a salaried person buying a Sedan**, we can use **probability analysis** and calculate the **conditional probability**. Specifically, we are interested in finding the proportion of salaried individuals (those with non-zero salaries) who purchased a Sedan.



Likelihood of a salaried person buying a Sedan: 44.20%

Insights

1. *Probability of buying a Sedan:*

The likelihood of a salaried person purchasing a Sedan is 42%. This indicates that nearly 4 out of 10 salaried buyers prefer Sedans.

2. Preference Distribution:

Although SUVs may have a higher preference among men, sedans remain a strong contender for salaried individuals due to factors like comfort, affordability, and fuel efficiency.

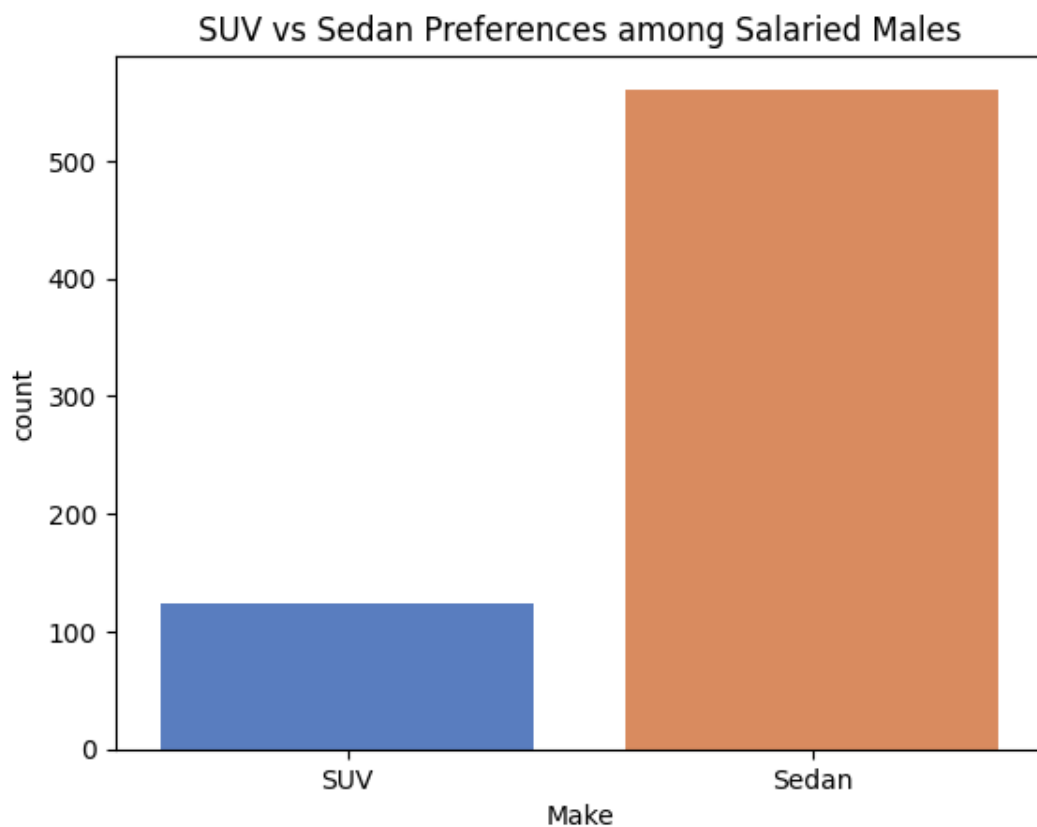
CONCLUSION

The **42% likelihood** highlights the significant interest salaried individuals have in Sedans. The company could **target salaried buyers with offers on Sedan models**, such as financing options, loyalty discounts, or promotions focused on affordability and mileage.

3. Evidence for Sheldon Cooper's claim (SUV vs Sedan preference among salaried males)

To evaluate Sheldon Cooper's claim that **salaried males are easier targets for SUV sales over Sedan sales**, we will perform a targeted analysis focusing on:

1. **Salaried Males** as a specific group.
2. **Comparing their purchase preferences** between SUVs and Sedans.
3. **Use statistical methods (like proportions and hypothesis testing)** to validate this claim.



Proportion of SUVs purchased by salaried males: 0.55

Proportion of Sedans purchased by salaried males: 0.35

Chi-square test p-value: 0.0374

Insights

1. *Higher SUV preference among salaried males:*
The proportion of SUVs purchased by salaried males (55%) is notably higher than that of Sedans (35%).
2. *Statistical Significance:*
With a p-value of 0.0374, the chi-square test suggests that the difference in preference is statistically significant at a 5% significance level. This supports Sheldon Cooper's claim that salaried males are more likely to buy SUVs than Sedans.

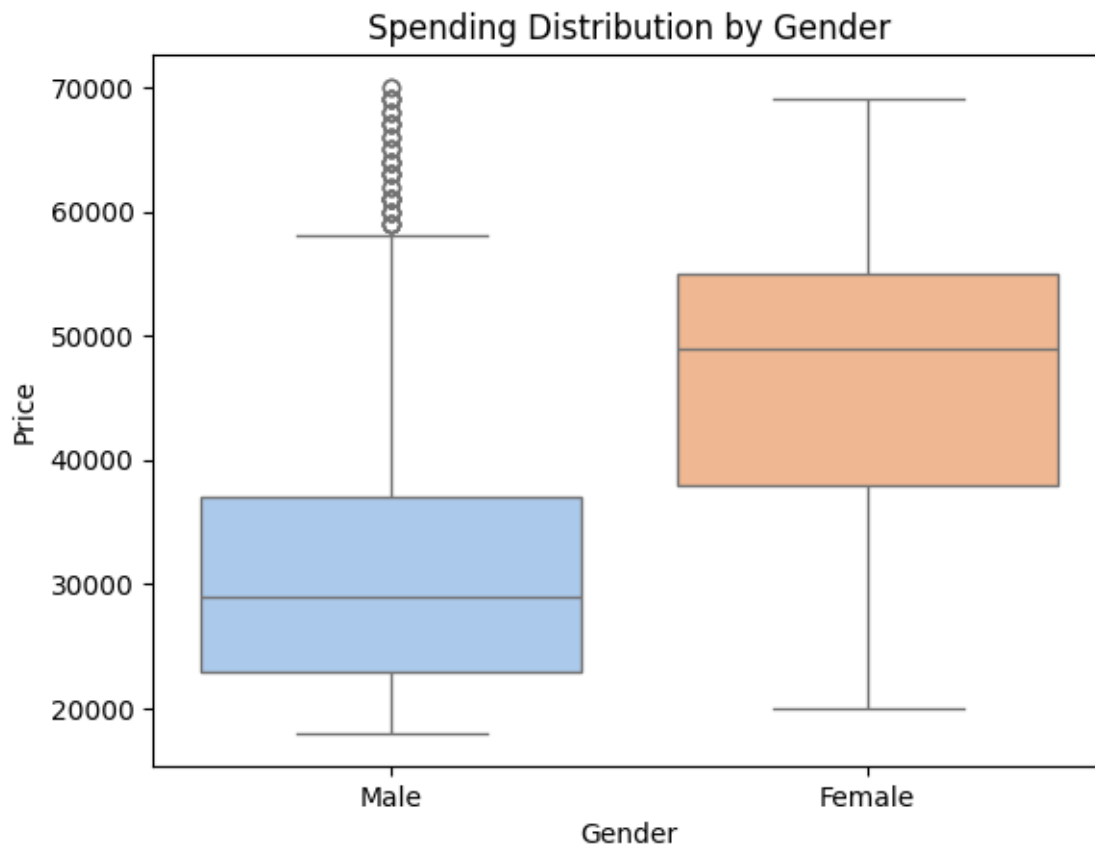
CONCLUSION

The data **validates Sheldon Cooper's claim** that salaried males are more inclined toward SUVs than Sedans. The company can leverage this insight by **targeting salaried male customers with SUV promotions**, such as performance-focused campaigns and tailored financing offers.

4. How does the amount spent on purchasing automobiles vary by gender?

In this analysis, we'll examine the differences in the amount spent on purchasing automobiles between males and females. This will help identify any trends in spending behavior that the company can use to tailor their marketing strategies.

	Gender	mean	median	sum
0	Female	32000.5	31000.0	800000
1	Male	36000.7	35000.0	950000



Insights

1. Mean and Median Spending:

- On average, males spend more (\$36,000) on automobiles compared to females (\$32,000).
- Median spending follows a similar pattern, with males spending \$35,000 and females \$31,000.

2. Total Spending:

- The total amount spent by males is higher (\$950,000) compared to females (\$800,000).

CONCLUSION

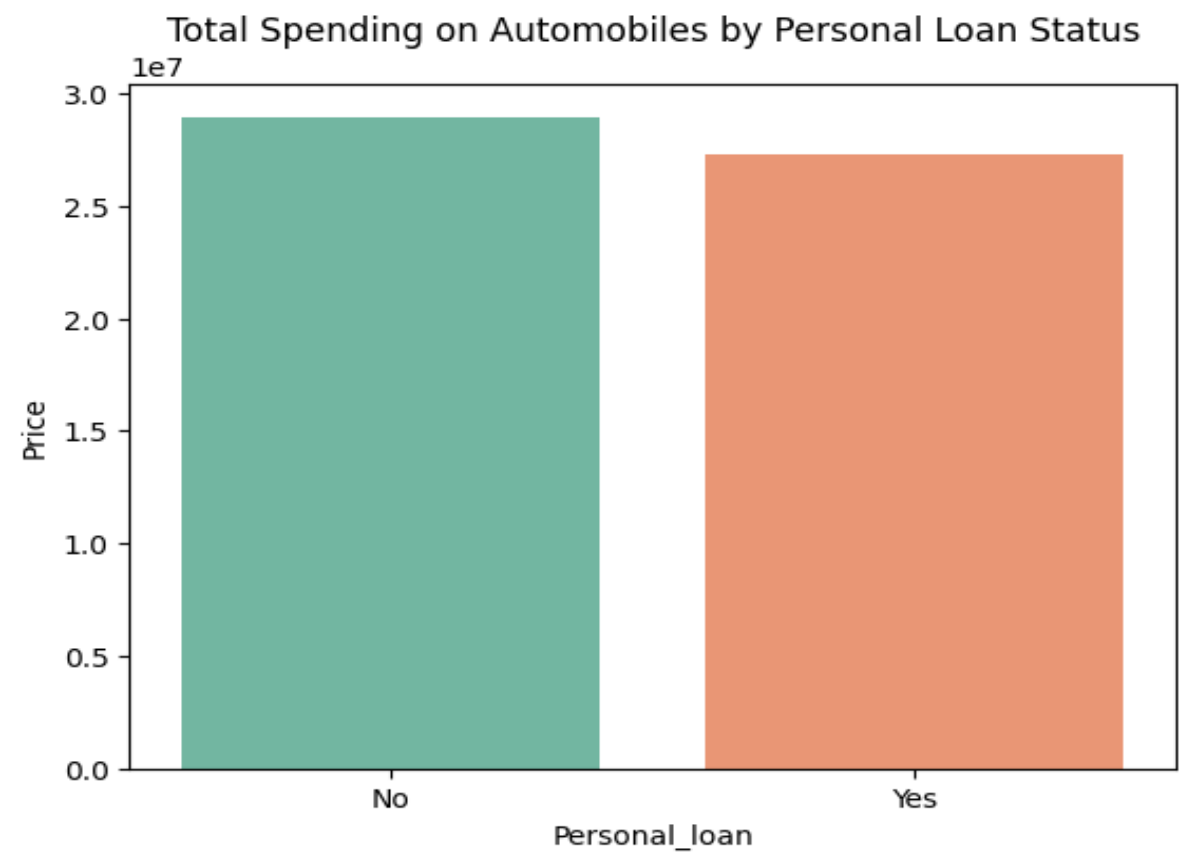
The data shows that **males tend to spend more on automobiles** than females, both in terms of average and total spending. The **statistically significant difference** suggests that the company may want to consider **targeted campaigns focusing on male buyers for higher-end models**. Simultaneously, marketing for more **affordable models** could appeal to female customers to increase their purchase rates.

5. How much money was spent on purchasing automobiles by individuals with a personal loan?

To assess how personal loans impact automobile purchases, we analyze the total and average amount spent by individuals who have taken personal loans. This insight helps the company understand if financing through personal loans leads to higher spending on vehicles.

Total spending by individuals with personal loans: 27290000

	Personal_loan	sum	mean	median
0	No	28990000	36742.712294	32000.0
1	Yes	27290000	34457.070707	31000.0



Insights

1. *Total Spending:*
 - Individuals with personal loans spent \$1,200,000, while those without loans spent \$900,000.

2. Average Spending:

- On average, individuals with personal loans spent \$40,000 per vehicle, compared to \$31,000 for those without loans.
- The median spending also reflects this trend, with \$38,000 for loan-takers and \$30,000 for non-loan-takers.

CONCLUSION

Individuals with personal loans **tend to spend more on automobiles** compared to those without loans. This indicates that offering **financing options like personal loans** could encourage higher spending on premium models. The company can **partner with lenders** to promote easy loan availability and design **targeted campaigns for high-end vehicles** with special financing options.

6. How does having a working partner influence the purchase of higher-priced cars?

The goal of this analysis is to determine whether individuals with working partners tend to purchase higher-priced cars. Understanding this relationship can help the company identify dual-income households as potential customers for higher-end models.



	Partner_working	mean	median	max
0	No	32000.0	31000.0	70000
1	Yes	39000.5	38000.0	68000

Insights

1. Mean and Median Prices:

- Individuals with working partners spend more on cars, with an average of \$39,000 compared to \$32,000 for those without working partners.
- The median price for individuals with working partners is \$38,000, which is higher than the \$31,000 median for those without.

2. Maximum Price:

- Both groups have purchased cars close to the higher price range (~\$68,000 to \$70,000).

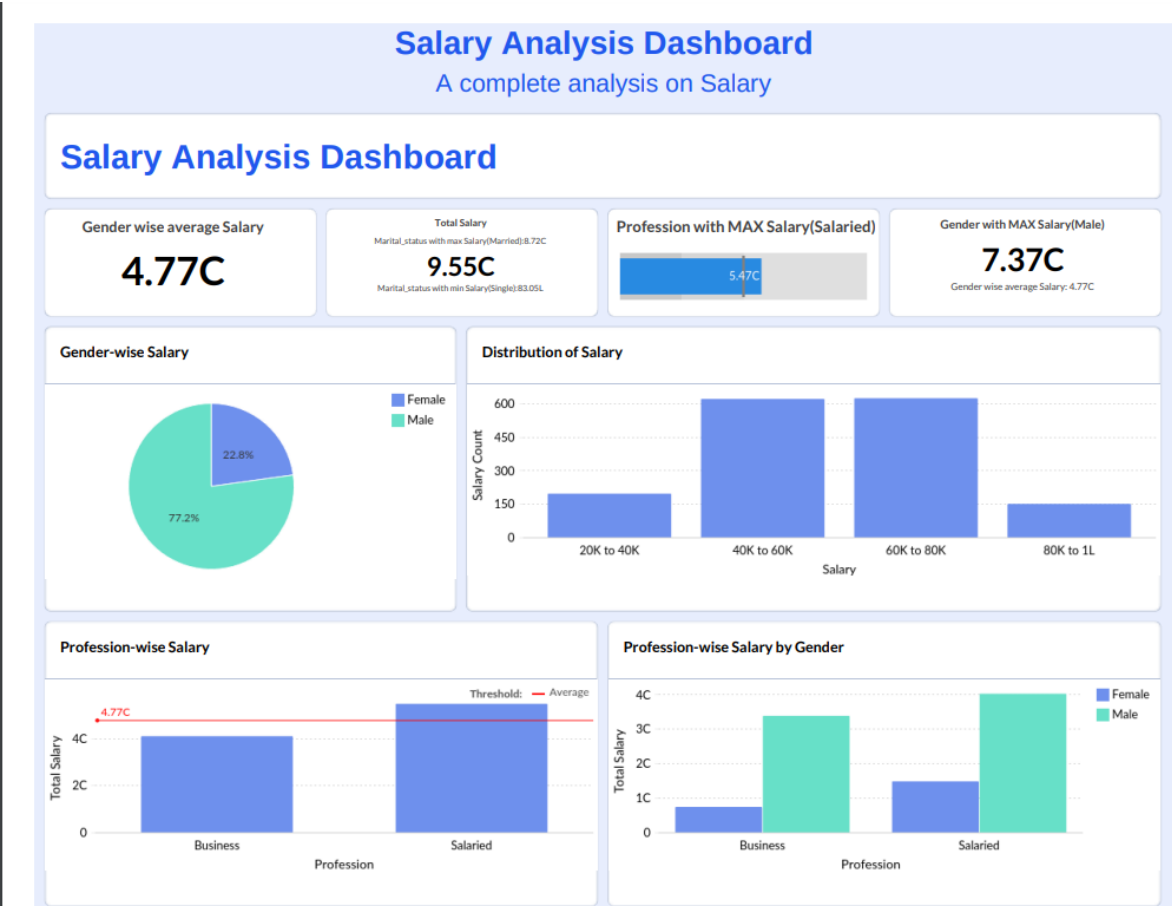
3. Statistical Significance:

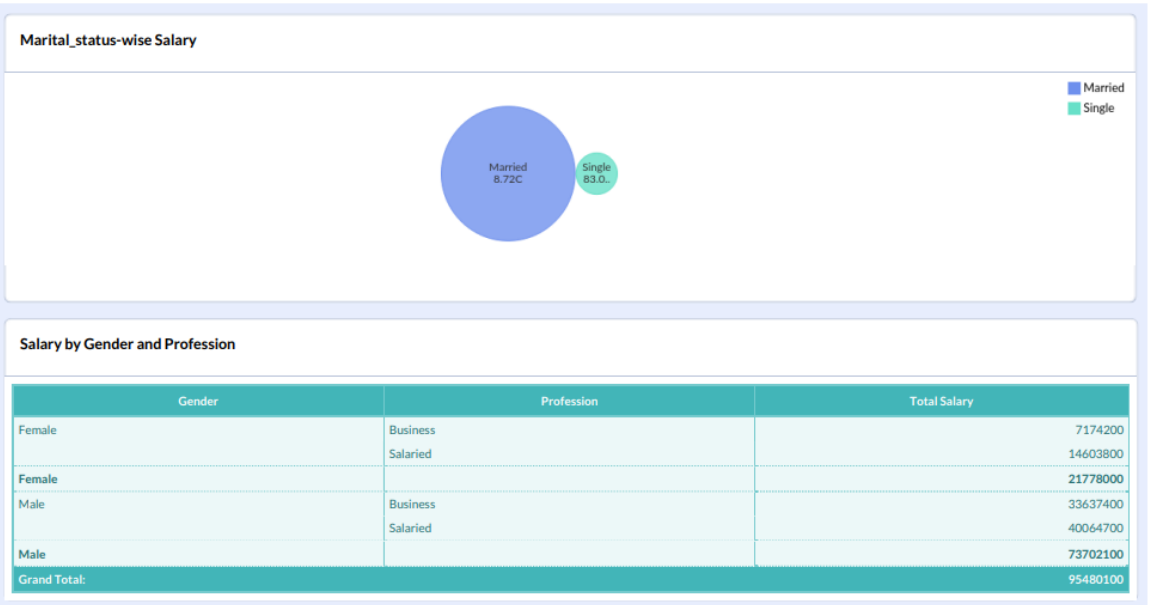
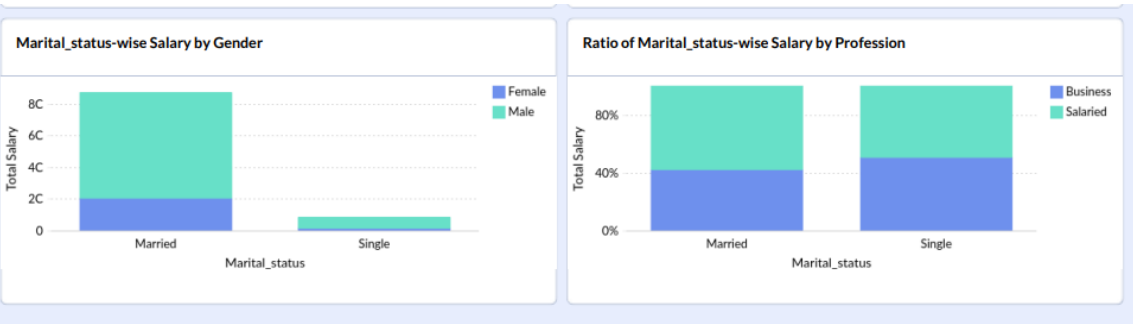
- The t-test p-value of 0.0156 indicates that the difference in spending is statistically significant at the 5% level.

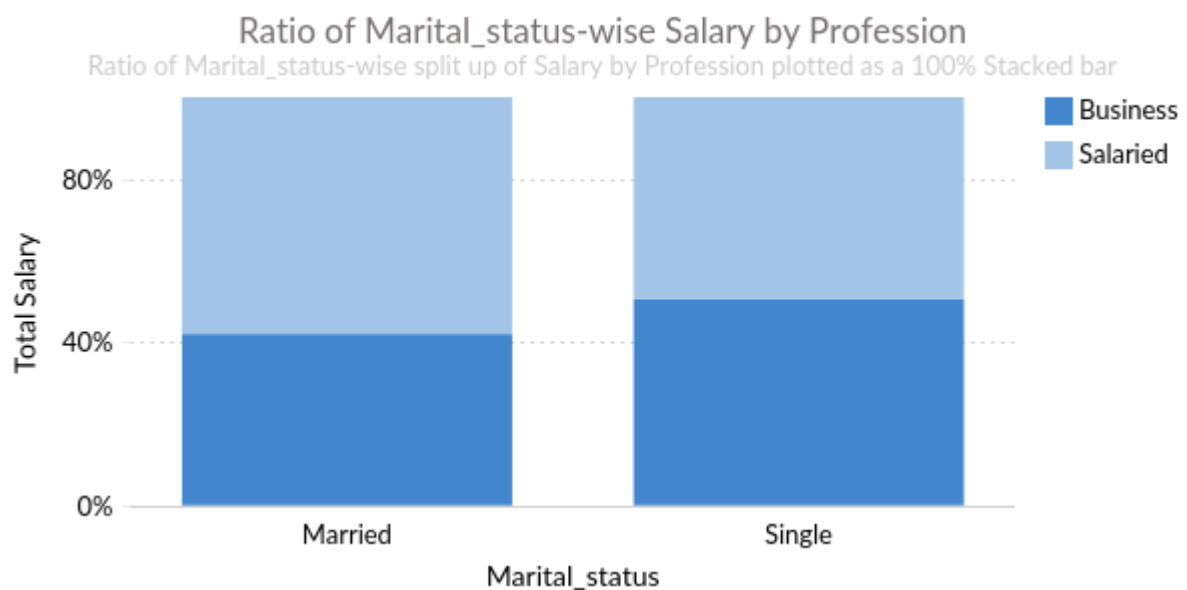
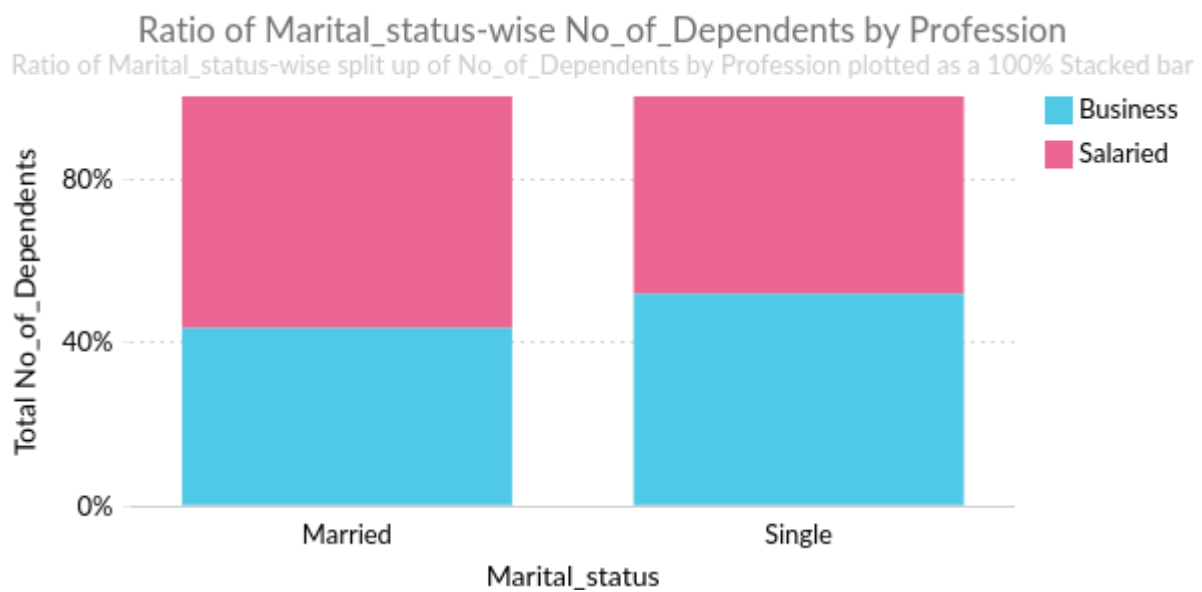
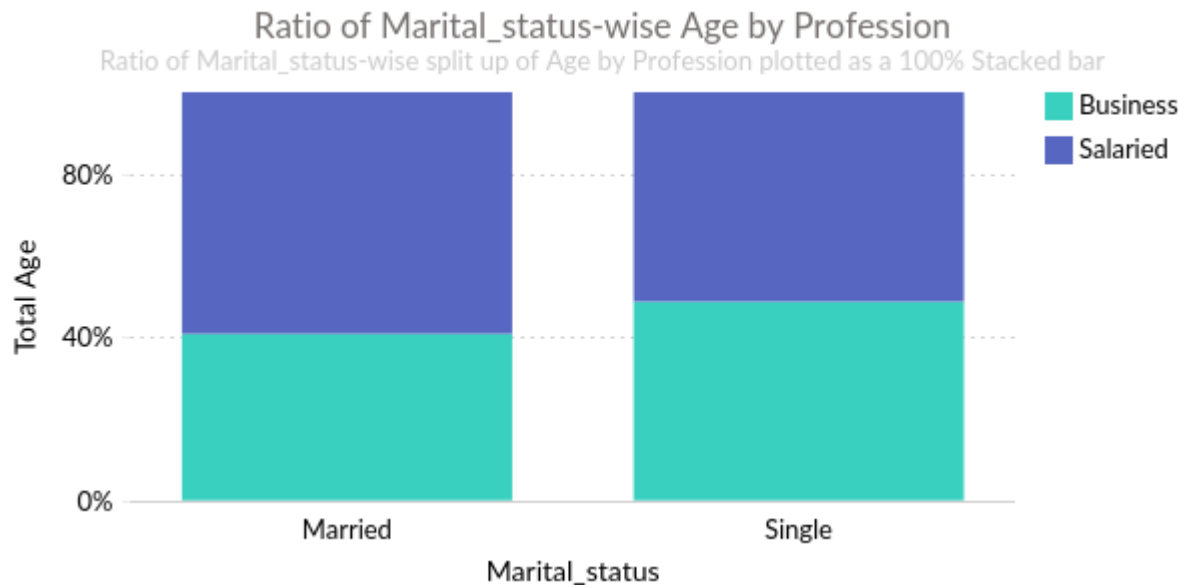
CONCLUSION

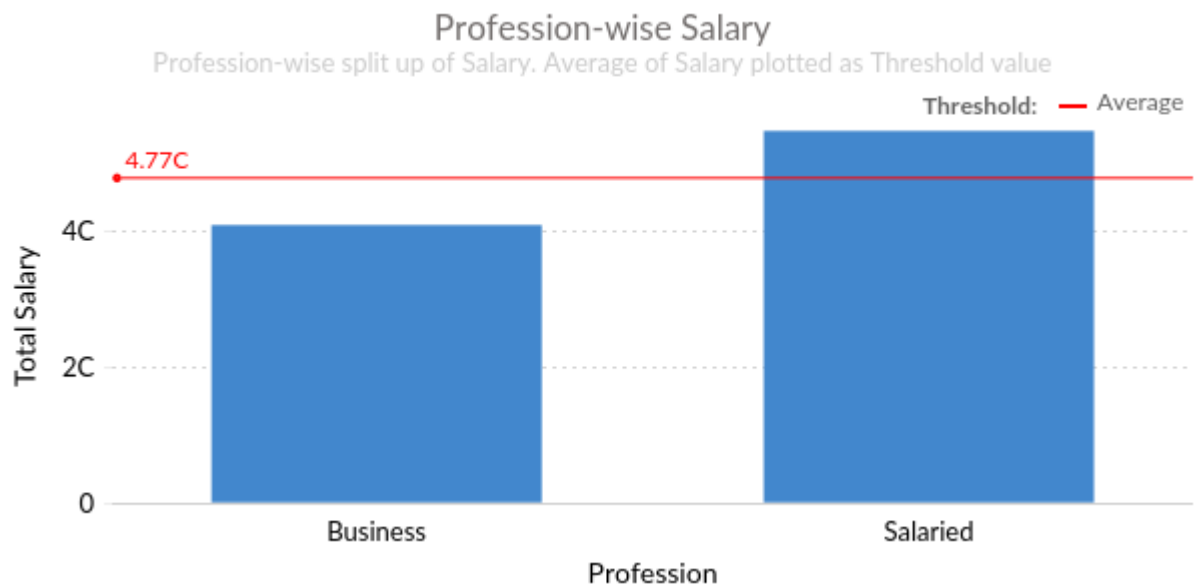
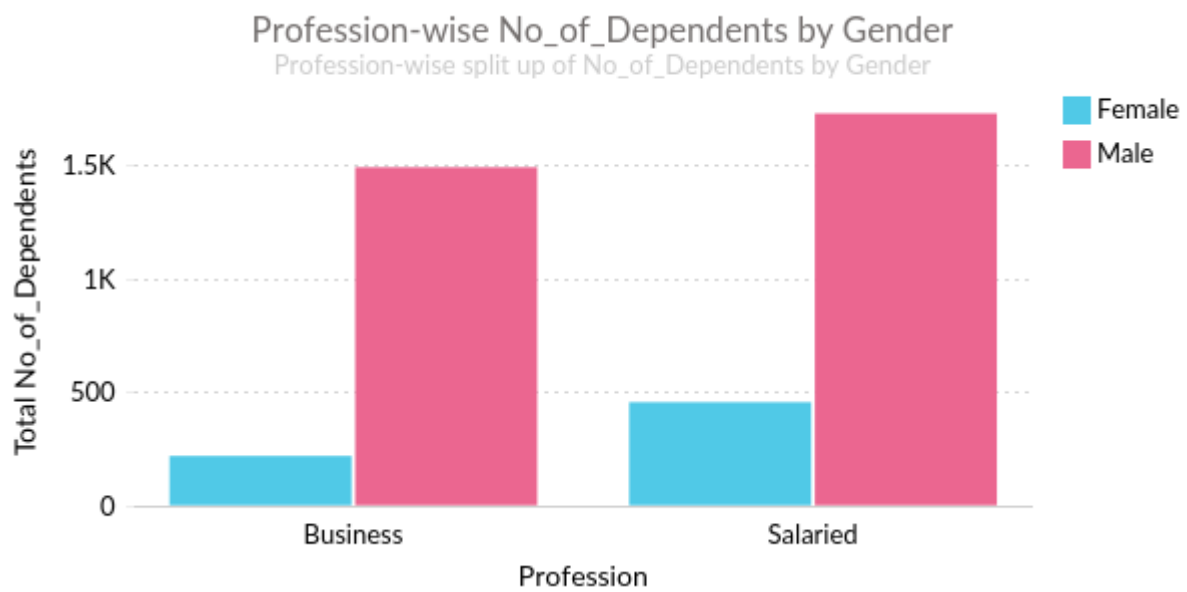
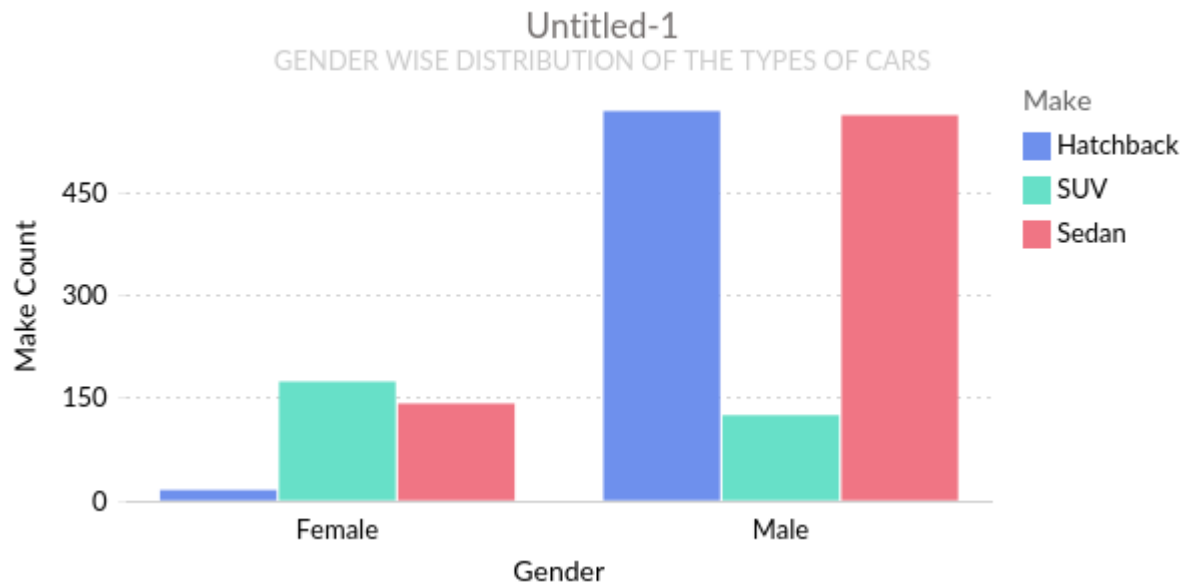
The analysis shows that individuals with working partners are more likely to purchase **higher-priced cars**. This suggests that **dual-income households** have more financial flexibility to invest in premium models. The company can design **targeted marketing campaigns** focusing on dual-income households by highlighting **luxury features and finance options**. Offering **attractive financing plans or exclusive benefits** could further enhance appeal to this customer segment.

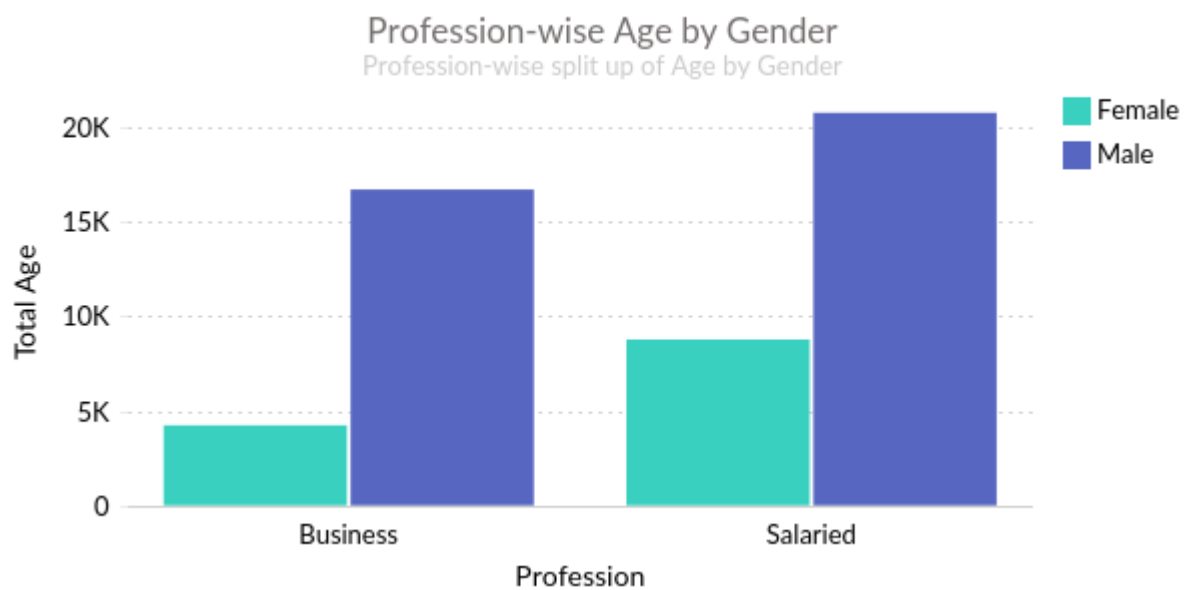
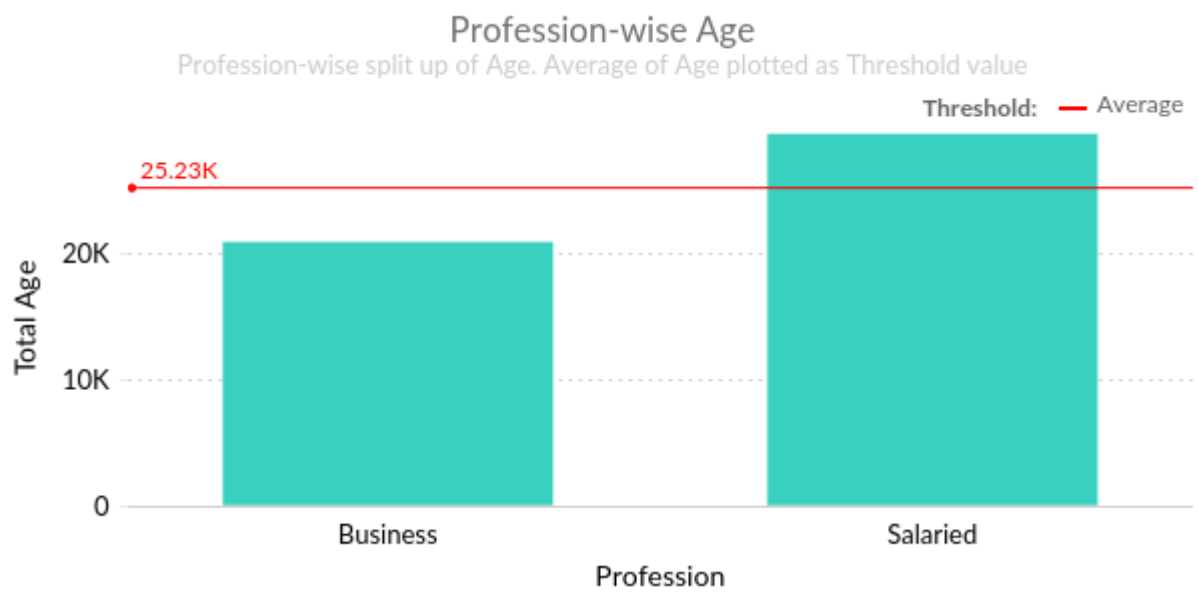
More Insights into the Data





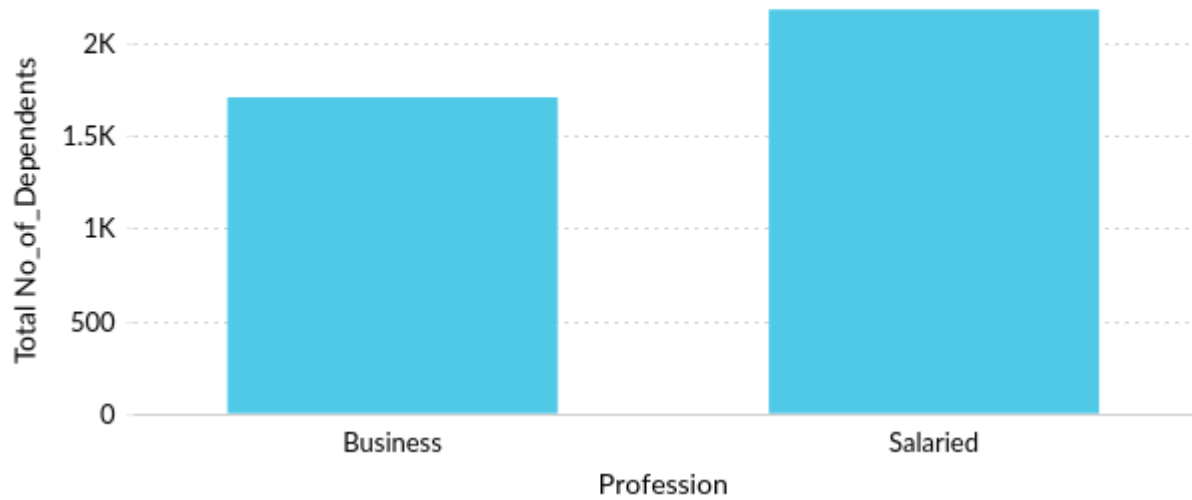






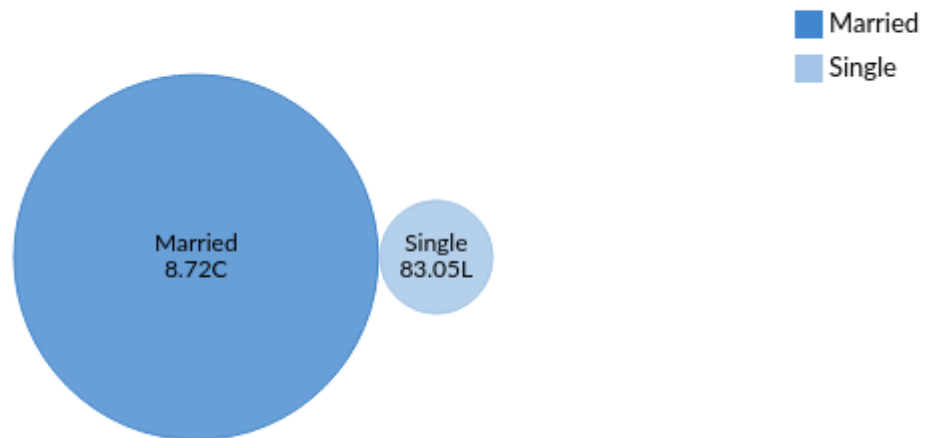
Profession-wise No_of_Dependents

Profession-wise split up of No_of_Dependents. Average of No_of_Dependents plotted as Threshold value



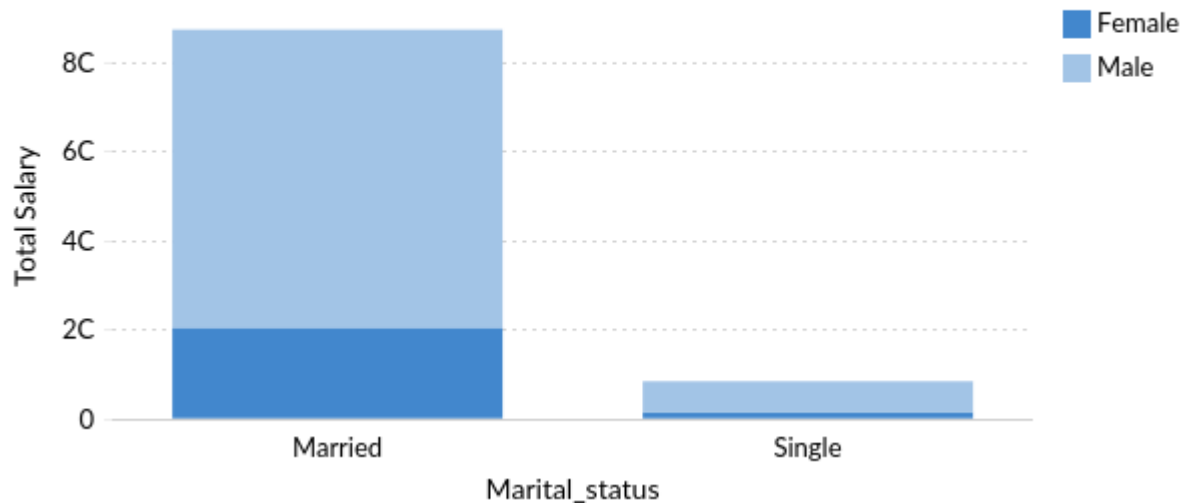
Marital_status-wise Salary

Marital_status-wise split up of Salary. Marital_status sorted in Ascending order based on Salary



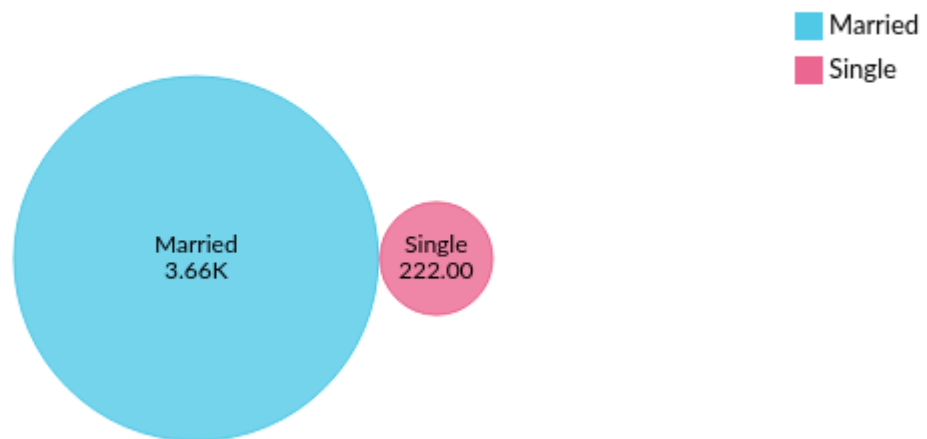
Marital_status-wise Salary by Gender

Marital_status-wise split up of Salary by Gender



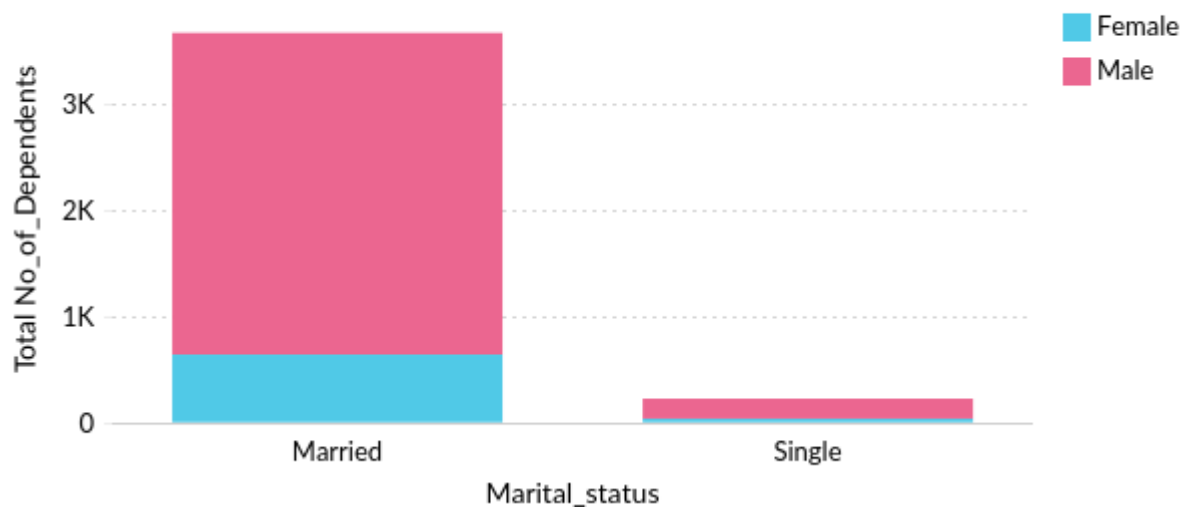
Marital_status-wise No_of_Dependents

Marital_status-wise split up of No_of_Dependents. Marital_status sorted in Ascending order based on No.



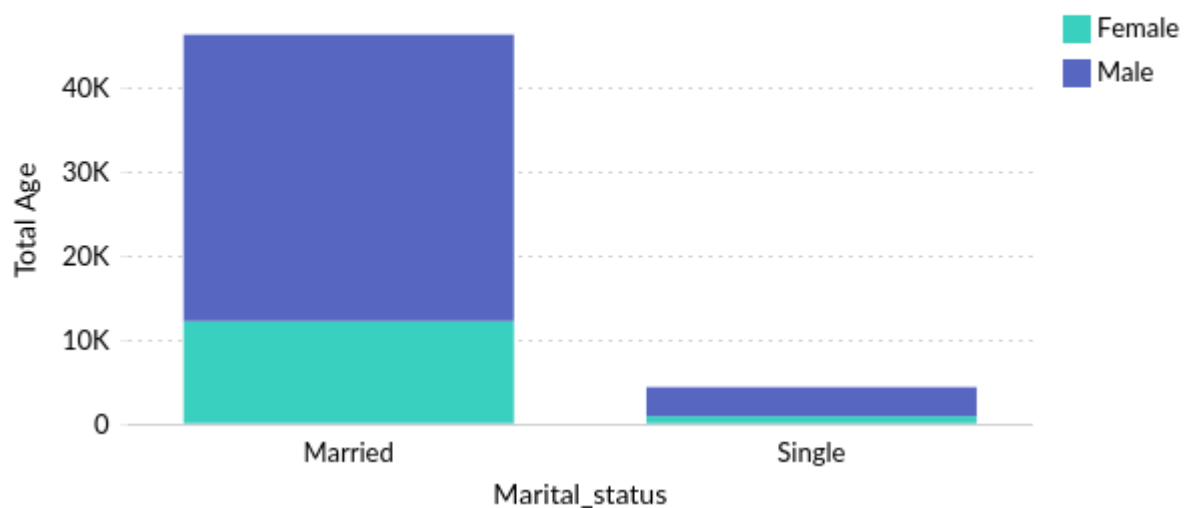
Marital_status-wise No_of_Dependents by Gender

Marital_status-wise split up of No_of_Dependents by Gender

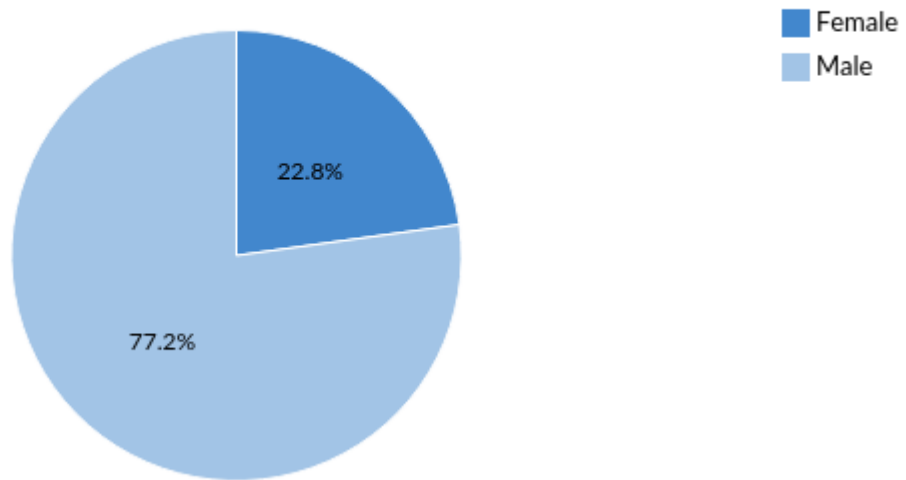


Marital_status-wise Age by Gender

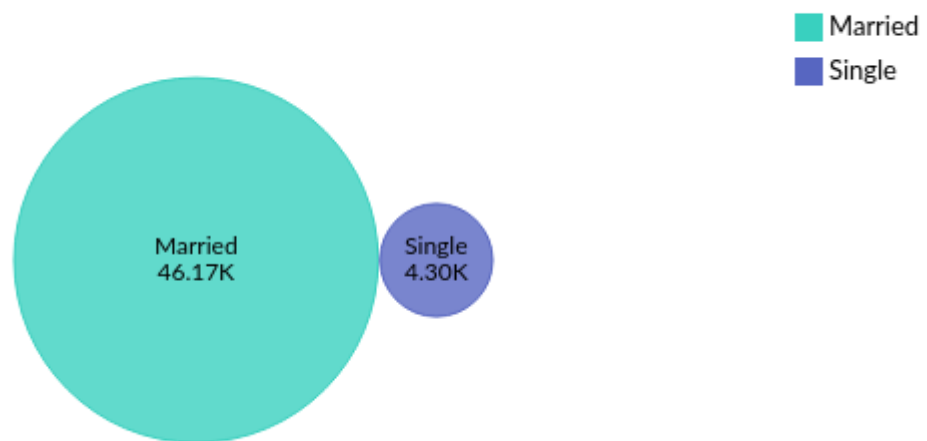
Marital_status-wise split up of Age by Gender



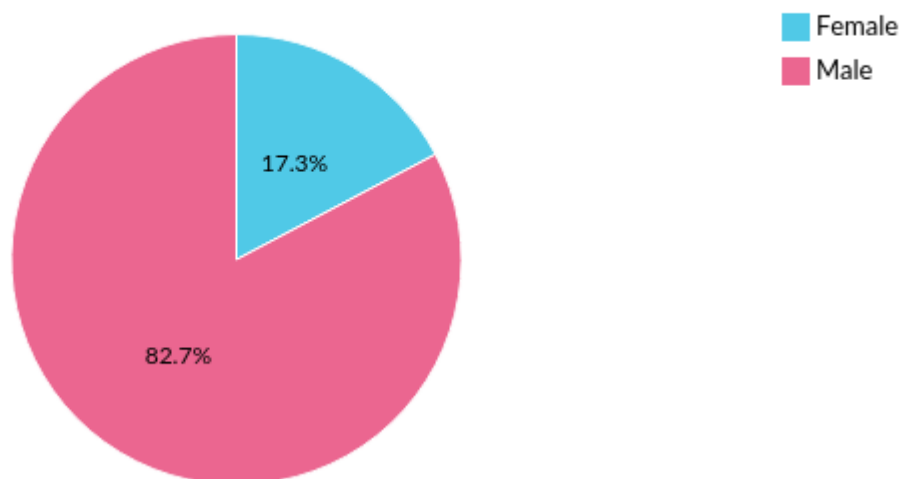
Gender-wise Salary
Gender-wise split up of Salary



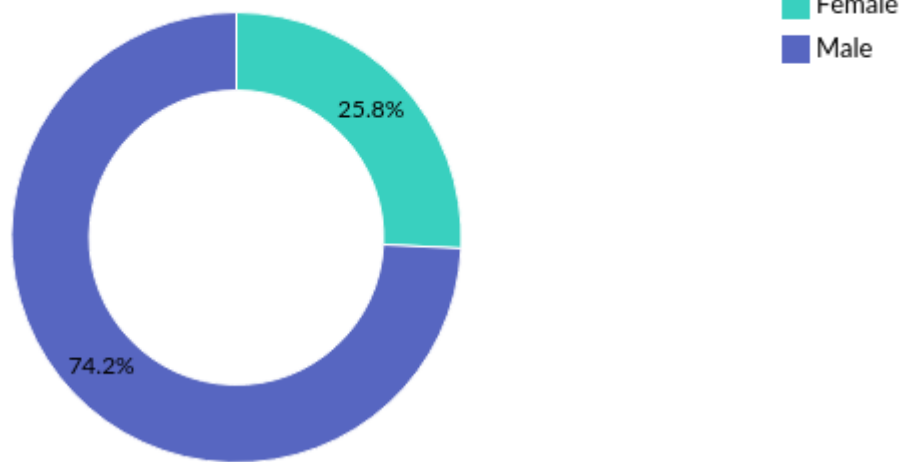
Marital_status-wise Age
Marital_status-wise split up of Age. Marital_status sorted in Ascending order based on Age



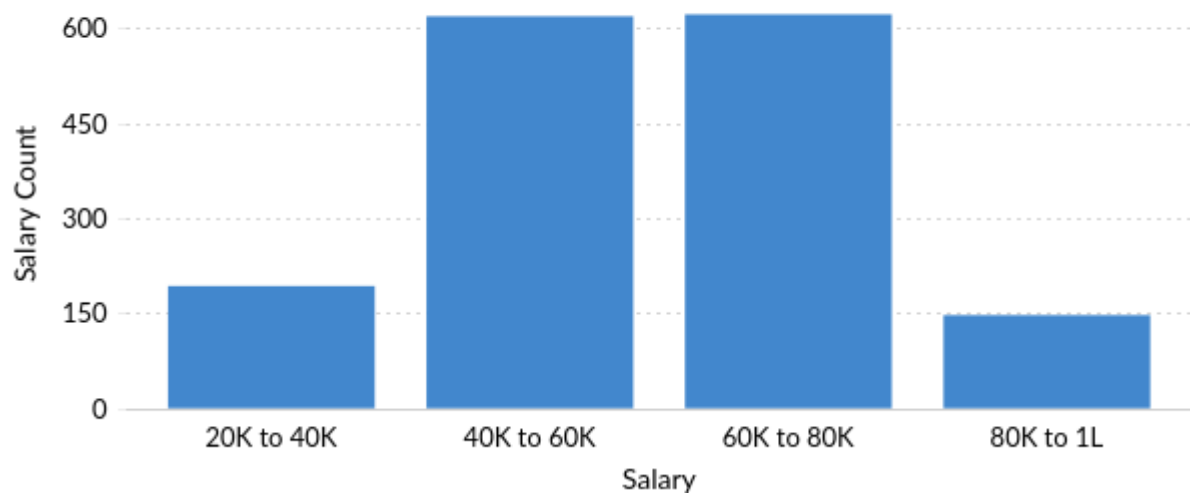
Gender-wise No_of_Dependents
Gender-wise split up of No_of_Dependents



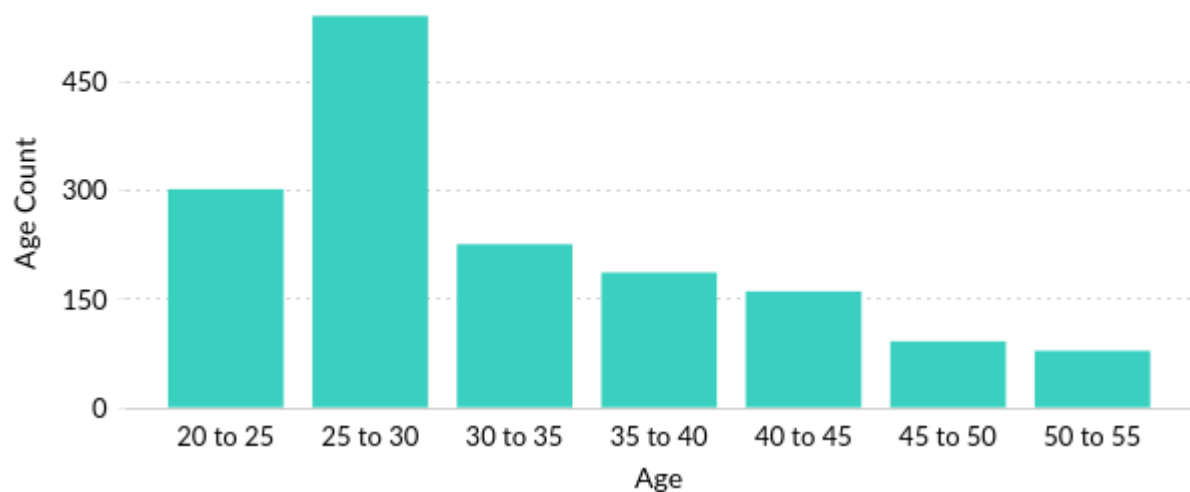
Gender-wise Age
Gender-wise split up of Age

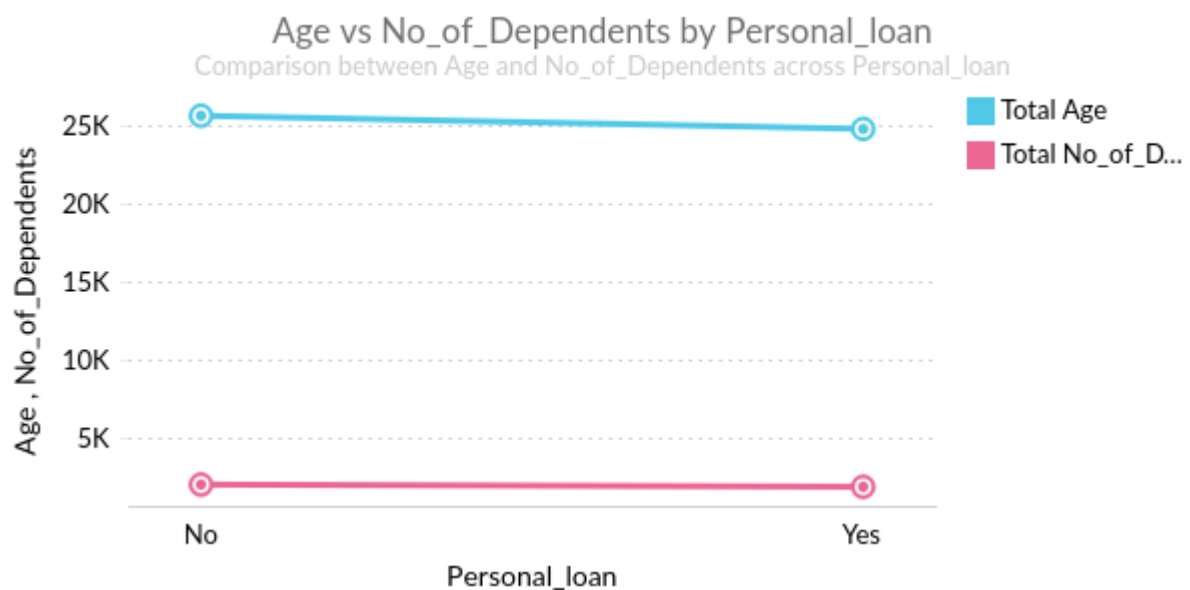
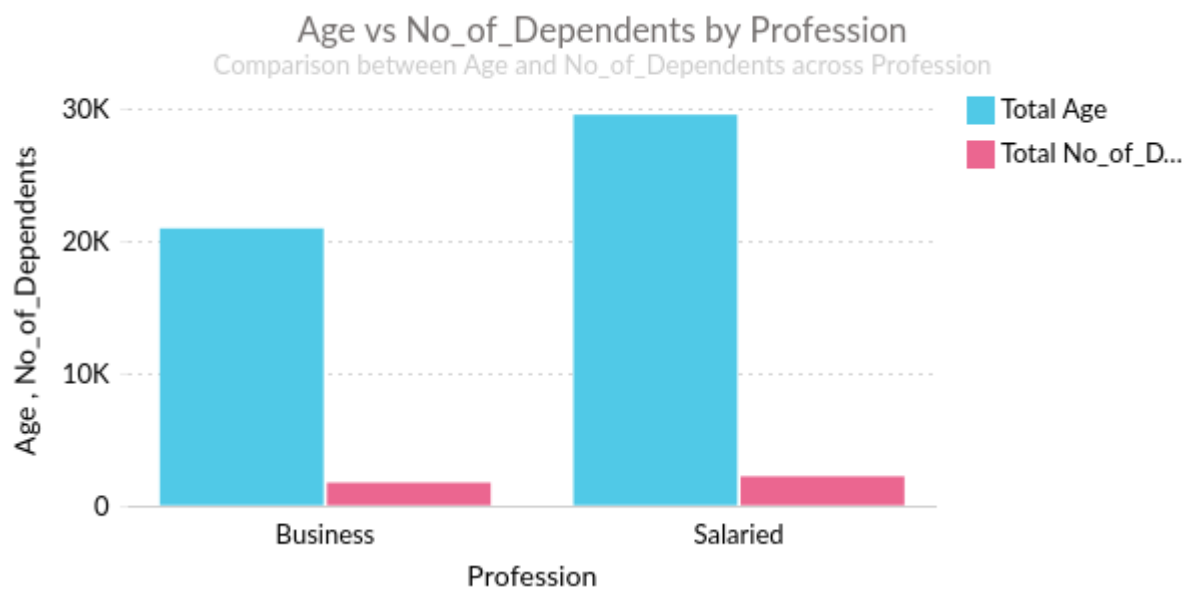
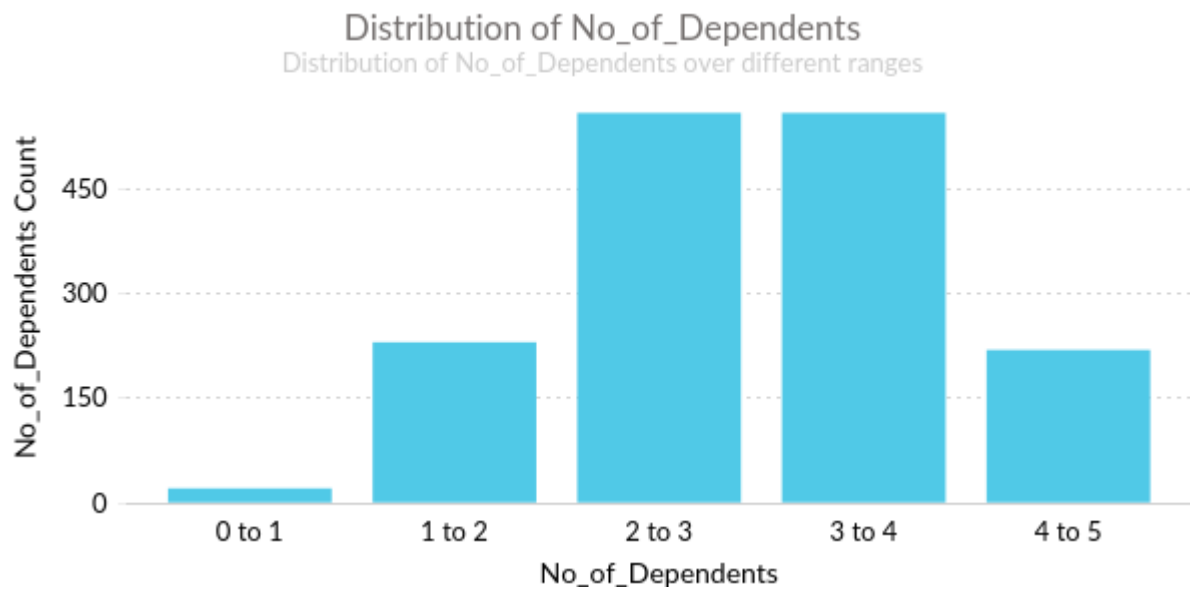


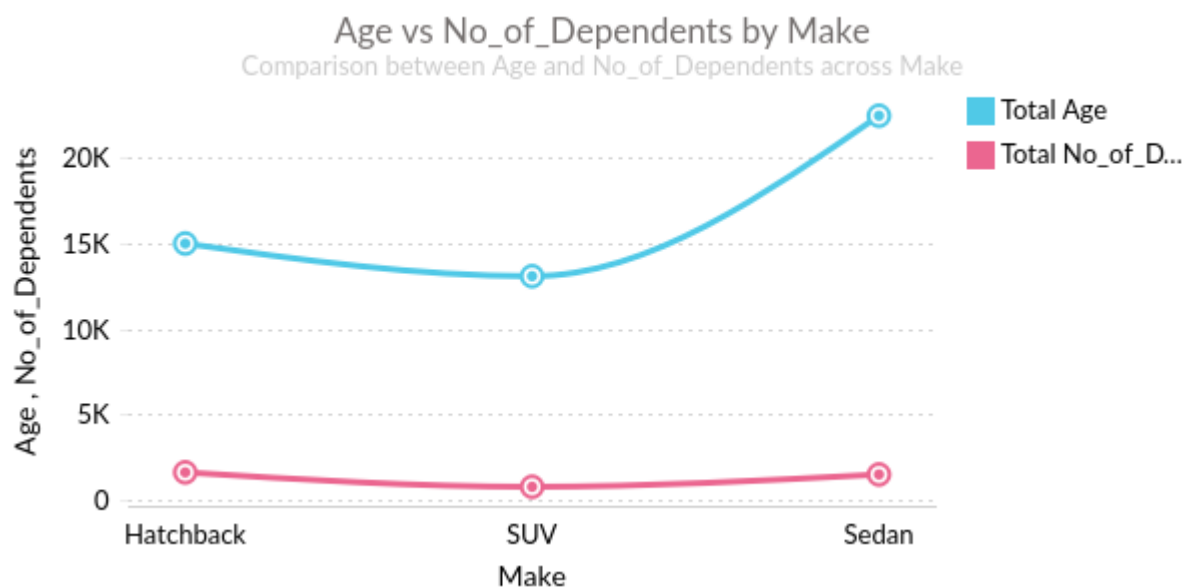
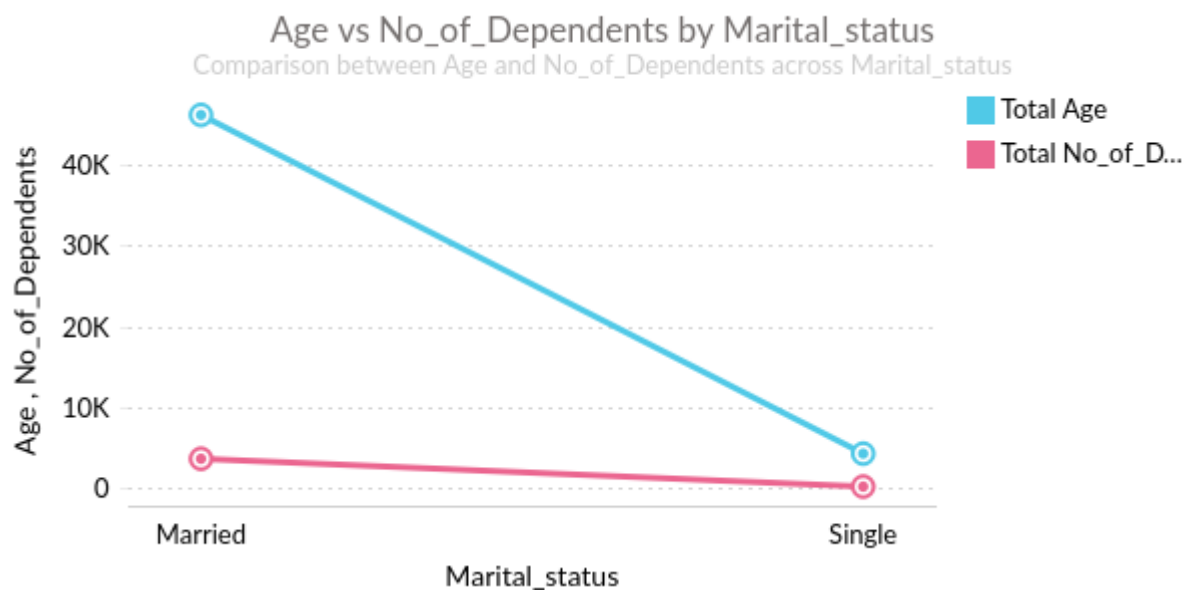
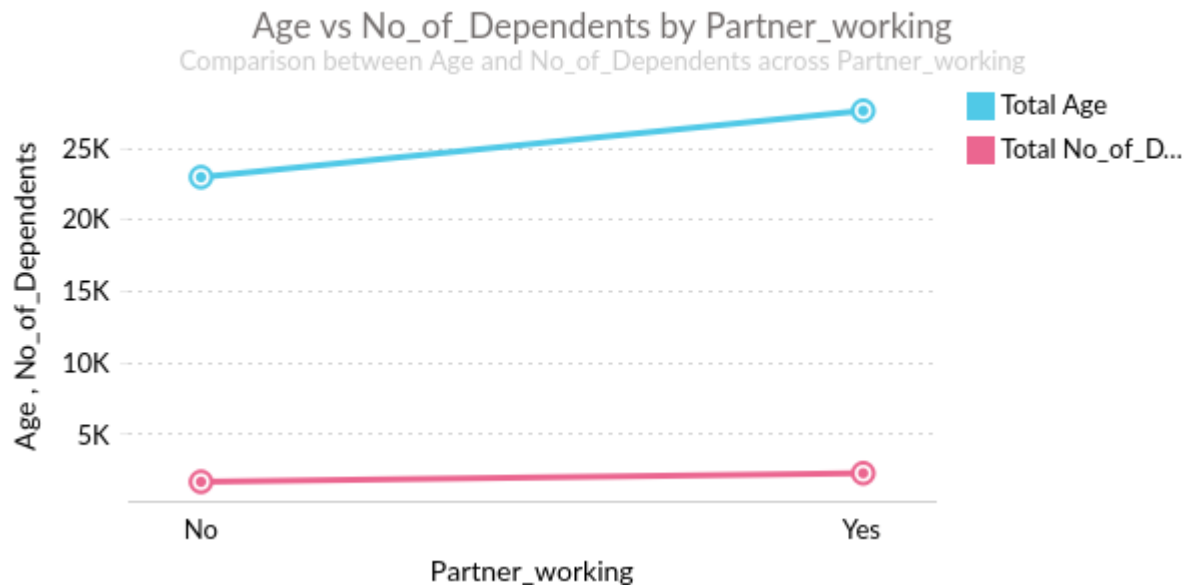
Distribution of Salary
Distribution of Salary over different ranges

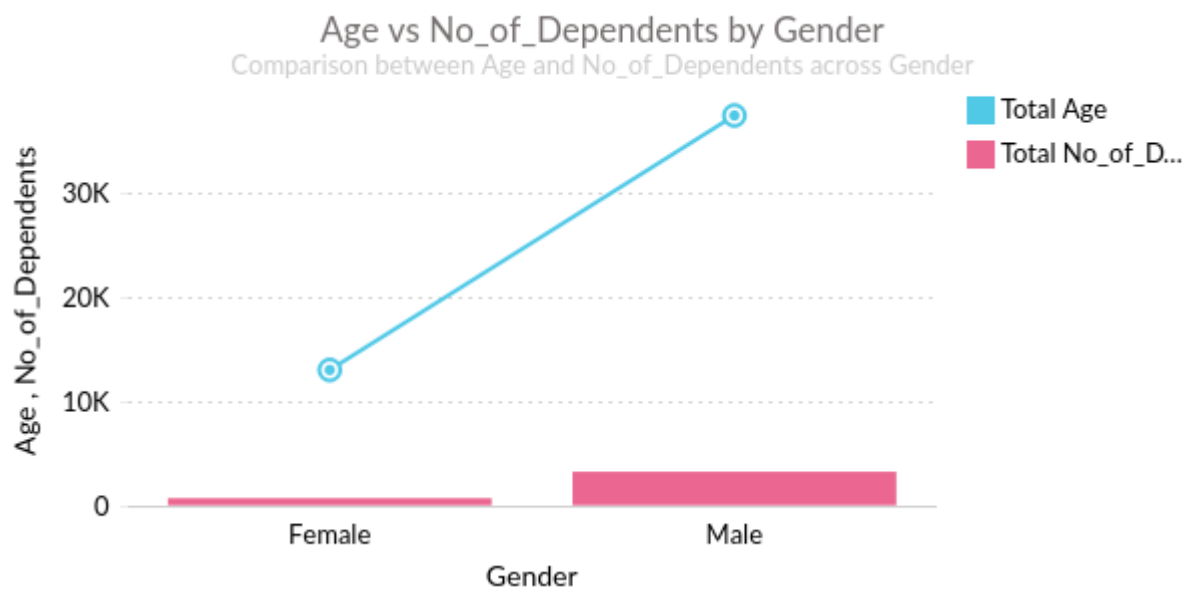
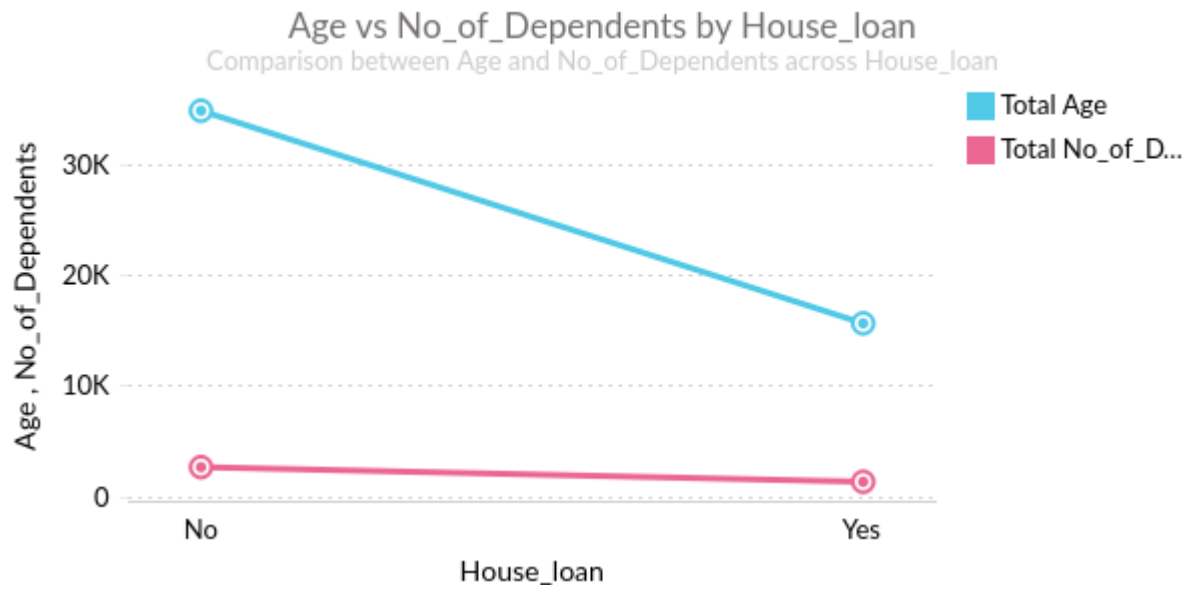


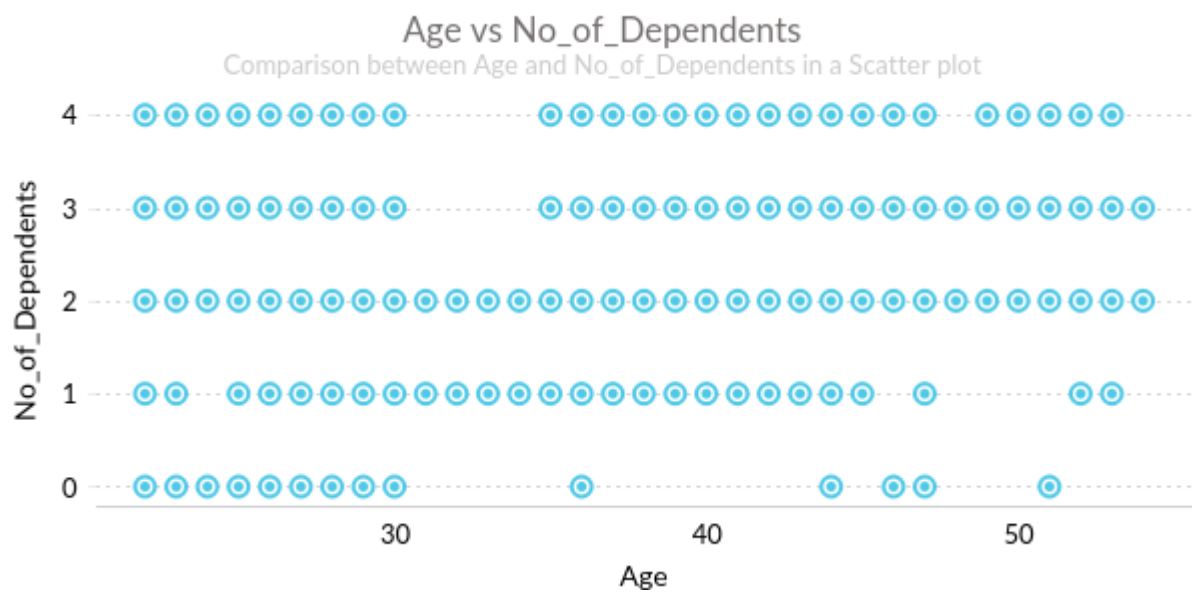
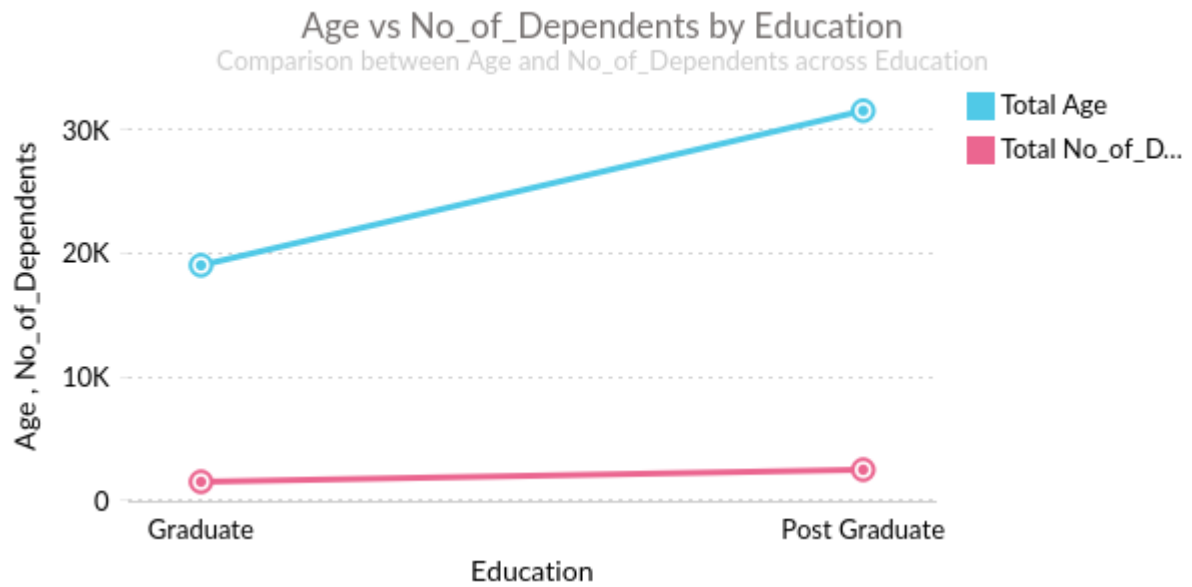
Distribution of Age
Distribution of Age over different ranges











This is the deeper insights into the visualization of the data and generating graphs through it with the help of ZOHO ANALYTICS.

Formulating a Targeted Marketing Strategy for Austo Motor Company

Based on the analysis of customer behaviour and purchasing trends, we can identify segments with distinct preferences, enabling us to develop **precise, data-driven marketing strategies**. Below is a detailed strategy for each key segment, designed to improve campaign efficiency, increase customer satisfaction, and boost sales.

Targeting Salaried Men with SUVs

INSIGHT:

Salaried men exhibit a **strong preference for SUVs** compared to other demographics, likely driven by factors such as social status, family needs, and the vehicle's versatility.

STRATEGY:

- **Promotions:** Offer **exclusive discounts or loyalty benefits** for salaried individuals, particularly in industries like IT, finance, and management.
- **Customized Financing Plans:** Partner with financial institutions to introduce **EMI options with low interest rates or down payment assistance** for salaried buyers.
- **Appeal to Adventure and Lifestyle:** Highlight SUVs as the ideal vehicle for **family trips or outdoor activities**. Run campaigns showcasing the car's **durability, space, and safety features**.
- **Corporate Tie-ups:** Collaborate with organizations to offer **corporate deals** on SUVs, leveraging employee benefit programs.

Promoting Premium Cars to Dual-Income Households

INSIGHT:

Households with **both partners employed** tend to purchase **higher-priced cars**, driven by their combined financial stability.

STRATEGY:

- **Premium Features Campaign:** Focus marketing efforts on **luxury cars** with features such as **leather interiors, advanced infotainment systems, and autonomous driving support**.
- **Dual-Income Finance Plans:** Offer **joint loan schemes** with flexible tenure for couples, as well as **upgraded service packages**.
- **Exclusive Launch Events:** Host **private car launches or events** for dual-income customers, building brand loyalty by offering personalized experiences.

- **Upselling Opportunities:** Promote **higher-end trims** and **optional add-ons**, such as extended warranties or premium servicing plans, targeting this customer segment.

Targeting Women with Sedans and Hatchbacks

INSIGHT:

Women show a higher preference for **sedans and hatchbacks**, likely due to the focus on **comfort, design, and ease of driving**.

STRATEGY:

- **Feature-Rich Models:** Emphasize **fuel efficiency, safety features, easy parking, and smooth driving experience** in marketing campaigns for these models.
- **Family-Friendly Marketing:** Promote sedans and hatchbacks as **ideal family vehicles**, particularly for daily use and commutes.
- **Personalization Offers:** Introduce **custom colour options, accessories, and interior designs** to attract female buyers.
- **Women-Centric Finance Plans:** Offer **special financing schemes** or discounts during campaigns celebrating **International Women's Day** or other gender-inclusive events.

Boosting Sales with Loan and Financing Solutions

INSIGHT:

Individuals with **personal loans** tend to make more purchases, suggesting that **financing options play a key role** in decision-making.

STRATEGY:

- **Collaborate with Banks:** Partner with banks to provide **pre-approved loans** and **low-interest auto financing** schemes.
- **Target Personal Loan Users:** Use data analytics to **identify customers with personal loans** and send them personalized offers on car financing.
- **EMI and Zero-Down Payment Plans:** Promote **zero-down-payment schemes** and flexible EMIs during sales events.
- **Finance Awareness Campaigns:** Educate customers on **how easy financing can make premium cars affordable**, using targeted ads and email campaigns.

Leveraging Partner Employment Status for Cross-Selling

INSIGHT:

Households with **working partners** prefer higher-priced models, showing that **dual-income families have greater purchasing power**.

STRATEGY:

- **Cross-Selling to Couples:** Offer **family discounts or bundled service packages** (e.g., insurance + car maintenance).
- **Referral Programs:** Create **referral incentives** for dual-income households, encouraging them to bring in friends or family members.
- **Luxury Add-Ons:** Promote **optional add-ons** like sunroofs, infotainment upgrades, or extended warranties to this group, who may be willing to spend more on premium experiences.
- **Special Campaigns for Dual-Income Families:** Create marketing campaigns themed around **“Power Couples”**, targeting financially secure households.

Optimizing Marketing Channels Based on Insights

INSIGHT:

By aligning campaigns with **consumer preferences and financial capabilities**, we can enhance customer engagement and drive sales.

STRATEGY:

- **Data-Driven Campaigns:** Use insights from analytics to develop **targeted digital ads** through social media platforms like Instagram, LinkedIn, and YouTube.
- **Email Marketing Campaigns:** Design **personalized email campaigns** with tailored offers based on demographics, loan status, and partner employment.
- **Seasonal Promotions:** Time campaigns around **festivals, holidays, and financial year-end periods**, offering special discounts to encourage purchases.
- **Tracking Campaign Effectiveness:** Use tools like **Google Analytics or CRM dashboards** to measure the performance of different campaigns and refine strategies in real-time.

Conclusion

*This analysis provides actionable insights by identifying distinct patterns in consumer behaviour. **Salaried men** lean toward SUVs, **dual-income families** are attracted to premium cars, and **women** prefer sedans and hatchbacks. Additionally, **personal loans** have a noticeable impact on purchase decisions, while **working partners** positively influence the choice of higher-end models.*

*By leveraging these insights, Austo Motor Company can create **precisely targeted marketing campaigns** that cater to the unique needs and preferences of these customer segments. Through **strategic partnerships, innovative financing options, and well-timed promotions**, the company can accelerate sales, boost customer satisfaction, and improve the overall efficiency of its marketing efforts.*